



Chinese medicinal materials' drying technologies advancements – Principles, energy performance, and influence on the bioactive components

Xin Li, Yunfeng Wang, Xun Ma, Ming Li, Decheng Kong, Phoodthahaksa Phavady, Qian Wang & Tarek Kh Abdelkader

To cite this article: Xin Li, Yunfeng Wang, Xun Ma, Ming Li, Decheng Kong, Phoodthahaksa Phavady, Qian Wang & Tarek Kh Abdelkader (2024) Chinese medicinal materials' drying technologies advancements – Principles, energy performance, and influence on the bioactive components, *Drying Technology*, 42:12, 1815-1845, DOI: [10.1080/07373937.2024.2394125](https://doi.org/10.1080/07373937.2024.2394125)

To link to this article: <https://doi.org/10.1080/07373937.2024.2394125>



Published online: 26 Aug 2024.



Submit your article to this journal [↗](#)



Article views: 523



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 24 View citing articles [↗](#)

REVIEW ARTICLE



Chinese medicinal materials' drying technologies advancements – Principles, energy performance, and influence on the bioactive components

Xin Li^a, Yunfeng Wang^{a,b,c}, Xun Ma^{a,b}, Ming Li^{a,b}, Decheng Kong^{a,b}, Phoodthahaksa Phavady^{a,b}, Qian Wang^{a,b}, and Tarek Kh Abdelkader^{a,d}

^aSolar Energy Research Institute, Yunnan Normal University, Kunming, China; ^bKey Laboratory of Solar Heating and Cooling Technology of Yunnan Provincial Universities, Kunming, China; ^cYunnan Provincial Rural Energy Engineering Key Laboratory, Kunming, China; ^dAgricultural Engineering Department, Fayoum University, Fayoum, Egypt

ABSTRACT

The distinctive pharmacological characteristics of Chinese medicinal herbs are rendering them increasingly popular as functional foods. However, fresh herbs are prone to mold and decay. Currently, drying processing is primarily employed to extend the storage time and effectively preserve the bioactive constituents of these herbs. After examining of the recent literatures on drying Chinese medicinal herbs, this paper provides a comprehensive overview of various drying technologies, including shade drying, hot air drying, solar drying, heat pump drying, microwave drying, far-infrared drying, vacuum drying, vacuum freeze drying, radio frequency drying, high voltage electric field drying and some combined techniques. While, an extensive illustration for each drying method principles, advantages/disadvantages, potential improvements and deep investigation of its energetic performance and the influence on the main active components in the drying process, such as polysaccharides, polyphenols, flavonoids, saponins, volatile oils, were summarized. The results showed that the combined drying technology not only had high efficiency and low energy consumption, but also preserved the bioactive medicinal components and sensory quality at the best. This paper provides a theoretical foundation and vast technical support for enhancing the value of Chinese medicinal herbs utilization and drying industry.

ARTICLE HISTORY

Received 6 June 2024
Revised 2 July 2024
Accepted 14 August 2024

KEYWORDS

Chinese medicinal materials; drying techniques; energy consumption; bioactive component; functional foods

1. Introduction

The development of Traditional Chinese Medicine (TCM) has been documented since the earliest known TCM text, Shennong Bencao Jing (The Classic of Materia Medica), during the Eastern Han Dynasty. This ancient medical system embodies a profound manifestation of Chinese civilization and wisdom. China holds a distinctive advantage in terms of TCM resources, encompassing approximately 12,000 medicinal plants.^[1,2] Presently, with the introduction and implementation of policies pertaining to TCM, the traditional Chinese medicine industry is experiencing a new phase of accelerated development, garnering increasing attention both in China and globally. Particularly during the COVID-19 pandemic, an abundance of data underscores the significant role played by TCM.^[3] Over the past decade, there has been a notable increase in the overall appeal of TCM's cultural value in China,^[4] with its evident efficacy when combined with Western medicine treatment.^[5,6]

Looking ahead, TCM holds immense significance and potential for development in both clinical practice^[7] and healthcare. Chinese medicinal herbs, as a vital component of TCM, play a pivotal role in the field. Abundant in bioactive compounds, which can be extracted through various methods. Those herbs are extensively employed for disease treatment and physiological regulation, constituting an integral element of TCM therapeutic interventions.

The majority of fresh herbs are susceptible to mold and decay, which hinders their optimal utilization, thereby necessitating the creation of specific preservation environments.^[8] Drying represents a fundamental procedure in the processing of Chinese medicinal herbs. It serves to eliminate a significant amount of moisture from the medicinal materials, thereby extending their storage lifespan, facilitating transportation, and preserving their essential medicinal properties to achieve optimal therapeutic effects. Consequently, primary processing of Chinese medicinal herbs has

garnered widespread attention.^[9] However, it's essential to recognize that most medicinal herbs are composed of intricate components with distinct physical and chemical properties, which diverge from those of other materials during the drying process. Drying Chinese medicinal herbs can lead to various outcomes, including the accumulation of secondary metabolites, stimulation of physiological metabolism, degradation of certain chemical components,^[10] and the loss of heat-sensitive medicinal compounds. Consequently, these factors result in alterations in the composition and content of active components such as polysaccharides, total phenols, flavonoids, saponins, volatile oils^[11] along with observable changes in color, shape, and taste before and after drying.^[12] The various outcomes resulting from the aforementioned drying process are not only relevant to Chinese medicinal herbs, but also warrant attention in any food drying procedures, such as the drying of vegetables, fruits, and agricultural products, as they directly impact the quality and utility of the final product.

To ensure the quality of medicinal herbs, it is imperative to employ an appropriate drying method tailored to the specific characteristics of each herb variety. Consequently, a plethora of drying techniques have been developed. Presently, primary drying methods for Chinese medicinal herbs encompass shade drying, hot air drying, solar drying, heat pump drying, microwave drying, far-infrared drying, vacuum drying, vacuum freeze drying, radio frequency drying, high voltage electric field drying, etc. Furthermore, various combined drying technologies have emerged, including microwave-far-infrared combined drying, heat pump-hot air coupling drying, hot air-microwave vacuum combined drying and so on. The continuous exploration of innovative drying technologies and the advancement of energy-efficient, environment-friendly, intelligent, and user-centric drying equipment signify pivotal trends in Chinese medicinal herbs drying. These endeavors aim to optimize efficiency and conserve energy throughout the entirety of the drying process.^[13–16]

Consequently, drawing upon the extant body of literature, this manuscript presents an exhaustive exposition on the intricacies inherent to the dehydration of Chinese medicinal materials. This synthesis amalgamates pertinent discoveries culled from a constellation of scholarly inquiries, spanning both national and international research endeavors. The aim is to elucidate the multifaceted challenges encountered in this domain and to underscore the significance of these findings in advancing the efficacy and quality of

medicinal herb drying methodologies. Firstly, it provides a comprehensive summary of current technologies, equipment principles, advantages and disadvantages, as well as the application status of drying methods for Chinese medicinal herbs. Subsequently, it highlights the significant potential for development in combined drying technology within this field. Lastly, it conducts an in-depth examination of the main components of Chinese medicinal herbs during the drying process. This review endeavors to establish a theoretical foundation and offer technical support for enhancing the utilization value of Chinese medicinal herbs, reducing equipment energy consumption, improving drying efficiency, investigating the drying characteristics of Chinese medicinal herbs, and providing insights for the comprehensive exploration and application of drying technology within the traditional Chinese medicine industry.

2. Drying technologies and equipment

During the drying process, temperature, relative humidity, air flow rate, and vacuum play pivotal roles in determining the quality of medicinal materials.^[17] Various physical and chemical changes occur to differing degrees throughout this process,^[18] particularly given the heat sensitivity of many medicinal materials. Improper selection of drying technology can result in the loss of essential medicinal components, ultimately compromising efficacy and quality.^[10] To date, drying methods have been broadly categorized into traditional and modern technologies, with the latter giving rise to various combined drying techniques.

2.1. Shade drying

Shading drying entails the natural drying of materials by allowing moisture to dissipate through wind in an environment shielded from direct sunlight. This method is characterized by low temperature conditions, resulting in a relatively prolonged drying process. Nevertheless, it effectively mitigated excessive volatilization of certain medicinal materials possessing volatile oil properties. Its notable feature lay in its non-thermal nature, rendering it suitable for drying heat-sensitive medicinal herbs.

The shade drying process involves placing the items directly in a darkened room to prevent exposure to light, and allowing for natural ventilation at normal temperatures during the drying process. Therefore, there is no mechanical drying facilities.

2.2. Hot air drying

The heat source was supplied by external conditions, and hot air was introduced into the drying chamber to facilitate convective heat transfer, thereby heating the material. Heat conduction occurred from the outside to the inside of the material. Throughout this process, the hot air removed moisture from the material's surface, serving as a drying medium responsible for carrying both heat and moisture.^[19] Simultaneously, adjusting the air humidity can help prevent surface hardening of the material to some extent.

The hot air drying method was favored for its ease of operation and low investment cost, making it widely utilized in the drying industry. However, this technique was not without its shortcomings. It tended to exhibit low efficiency, extended time consumption, excessive loss of medicinal components, and compromised sensory quality. As a result, it was considered unsuitable for the drying of Chinese medicinal herbs with high medicinal value.

2.3. Solar drying

2.3.1. Direct drying

The direct solar drying experimental device depicted in Figure 1 is a simple natural convection solar dryer. Constructed from aluminum material, the setup comprises a frame, tray and glass cover. Air enters the dryer's end and effectively removed moisture from the material through the thermal siphon effect. Simultaneously, the expelled hot air reduced humidity within the drying room, thereby accelerating the drying rate.^[20] In addition, the air movement inside the solar dryer can be categorized into active drying and passive drying. The structure of a passive solar dryer is simple, as the air in the dryer moves naturally due to buoyancy. For example, in a traditional solar

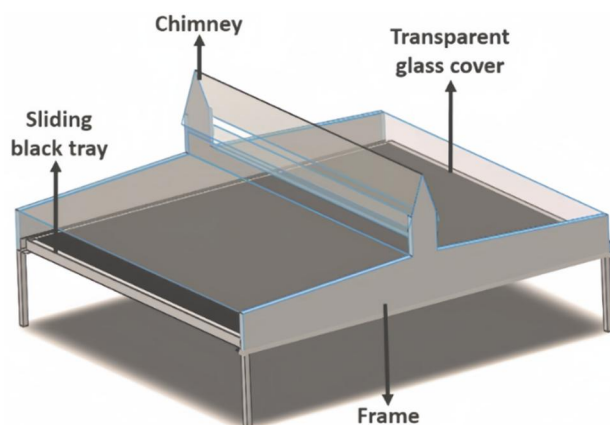


Figure 1. Schematic diagram of the direct solar dryer.^[20]

chimney dryer, hot air rises to the top of the chimney and then dries materials placed perpendicular to the bottom of the chimney, effectively improving the drying rate.^[21] The active dryer is equipped with a fan to enhance the air flow in the drying room and improve drying efficiency. In comparison to natural convection dryers, forced convection dryers significantly reduce energy consumption and shorten the drying cycle.^[22]

Importantly, it is noteworthy that the impact of light extends beyond the decomposition of pigments in medicinal materials; it also leads to photodegradation of phenols, flavonoids, monoterpenes, and other components present in these materials.^[10] For instance, light exposure can trigger strong photodegradation reactions in ferulic acid found in *Angelica sinensis*, resulting in a significant reduction in its content.^[23] Additionally, ketones in *Salvia miltiorrhiza* are susceptible to photodecomposition. Under conditions that minimized exposure to light, such as shade drying or hot air drying, the content of *tanshinone II A* significantly increased with higher drying temperatures.^[24] Therefore, when selecting drying methods, it was crucial to consider the presence of light-sensitive substances in Chinese medicinal herbs.

2.3.2. Indirect drying

In comparison to direct drying methods, indirect drying offered several advantages, including precise control over temperature, humidity, air velocity and the ability to prevent decomposition of certain active components due to exposure to light. Indirect drying typically involved a solar air heater and a drying chamber. Here, the solar air heater efficiently converts solar radiation into heat energy, achieving an efficiency exceeding 80%. The hot air generated at the outlet side of the collector was then directed into the drying chamber. Within the chamber, convective heat transfer occurred between the hot air and the materials, facilitating the drying process.^[25]

In addition to temperature, humidity, air velocity, solar radiation is the primary factor that impacts the performance of a solar dryer, thereby influencing the outlet temperature of the heater. Additionally, as the air flow rate increases, the difference between the outlet and inlet temperature of the heater decreases.^[26,27] Capturing enough solar radiation helps to speed up the flow of hot air, thereby increasing the drying rate, and solar radiation depends on collector performance. Therefore, it is essential to select an appropriate collector based on the type of material being dried.^[28]

2.3.3. Mixed-mode drying

To address the challenge of limited exposure to sunlight, various hybrid solar dryers and thermal energy storage dryers have been developed. Lakshmi et al.^[29] introduced a novel category of a mixed-mode forced convection solar dryer with a thermal energy storage system, as illustrated in Figure 2. Their findings revealed a significant time-saving of 60.7% when utilizing the hybrid solar dryer compared to the direct solar dryer. Additionally, it was observed that the hybrid solar dryer exhibited superior preservation capabilities for the total flavonoids content in *turmeric*. Specifically, the total flavonoids content in samples dried using the direct and hybrid methods were measured as 7.09 mg/g and 7.59 mg/g, respectively.

The design of a mixed-mode solar dryer is primarily based on the following aspects: (1) maximizing the utilization of solar energy resources by implementing a well-designed collector, drying chamber, and heat transfer system to convert solar energy into heat for the drying process. (2) Considering the specific characteristics and requirements of the materials being dried, such as material type, moisture content, and drying temperature, in order to ensure that the desired drying effect is achieved. (3) The performance can be realized by thermal analysis of mainly two parameters: collector outlet temperature and no-load

performance index (NLPI) (4) Taking into consideration the operational stability and safety of the equipment, meticulous selection of the most suitable heat storage materials and structures is undertaken to ensure the stable and efficient operation of the equipment. This ensures that there are no safety hazards during use.^[30]

Firstly, solar energy represented a clean and renewable source of power, making solar drying an environmentally friendly option that significantly reduced energy consumption and carbon emissions. Secondly, solar drying equipment featured a simple design and offered broad applicability, guaranteeing safety and reliability for large-scale drying operations within the Chinese medicinal herbs industry. According to China's classification standards for total solar energy resources, in 2022, most of Xinjiang, Tibet, Northwest and western Southwest regions, as well as most of Inner Mongolia, North China, central China, East China and South China have a total annual horizontal radiation amount exceeding 1400 kWh/m². Among these areas, the majority of Tibet and the central and northern parts of Qinghai possess the highest solar energy resources with a total annual horizontal irradiance exceeding 1750 kWh/m². Additionally, most of Xinjiang, a large portion of Inner Mongolia, central and eastern Northwest regions, North China, East

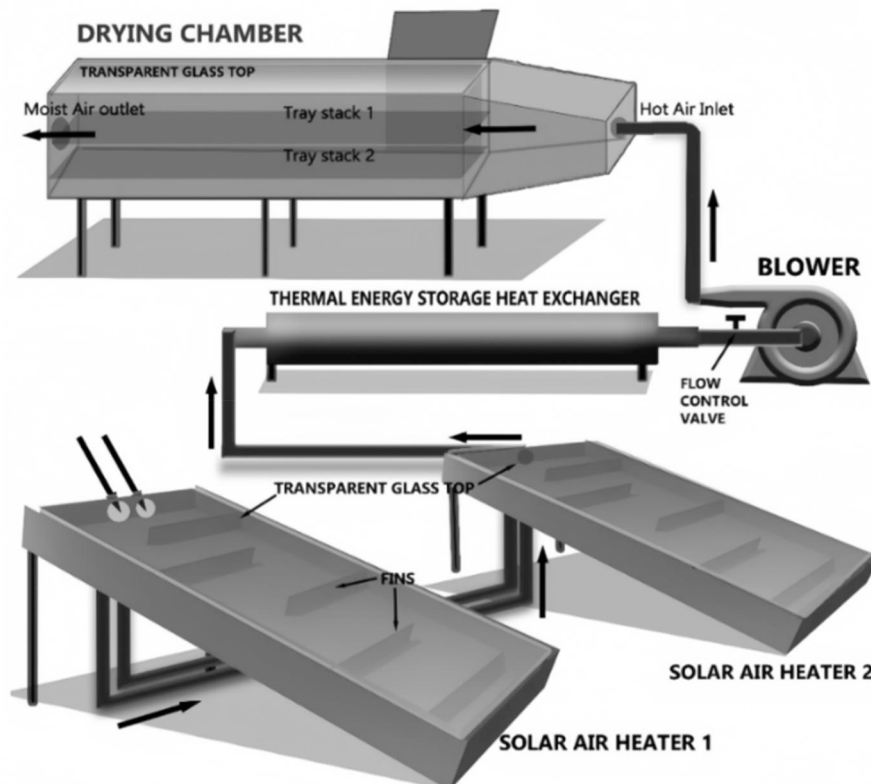


Figure 2. Schematic of mixed mode forced convection solar dryer with thermal energy storage system.^[29]

China and eastern South China also exhibit a high level of solar energy resources with a total annual horizontal radiation ranging from 1400 kWh/m² to 1750 kWh/m². Furthermore, the eastern Northeast region along with eastern Sichuan Chongqing Guizhou western Hunan northern and eastern Guangxi are considered rich areas for solar energy resources due to their total annual horizontal radiation amount ranging from 1050 kWh/m² to 2140 kWh/m². In fact, the total annual horizontal irradiation amount in all regions across China is almost always greater than 1050 kWh/m², thus there are essentially no general areas lacking in solar energy resources.^[31] This provides an excellent foundation and significant potential for the development of the solar energy drying industry in China.

2.4. Heat pump drying

The heat pump drying system primarily comprised of a compressor, condenser, throttle valve, evaporator, and other essential components. The entire process

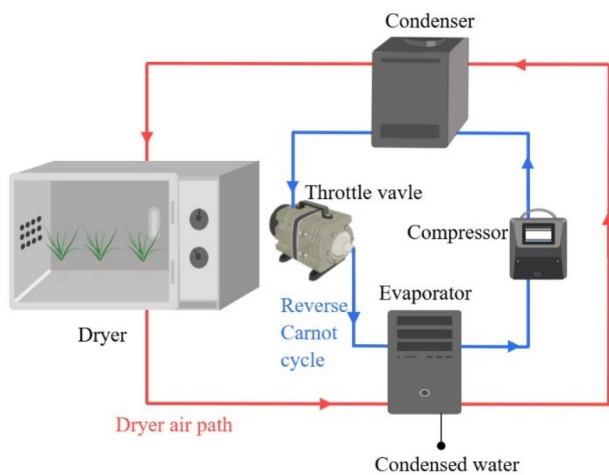


Figure 3. Schematic of heat pump dryer system.^[32]

adhered to the fundamental principles of the reverse Carnot cycle. The working medium in the pump was initially compressed into a high-temperature and high-pressure gas refrigerant, which then entered the condenser for liquefaction and condensation, releasing a significant amount of heat energy. Subsequently, the liquid refrigerant passed through the throttle valve where it undergoes throttling to become a low-temperature and low-pressure liquid. It then entered the evaporator as a low-temperature and low-pressure gas before finally reentering the compressor to complete the cycle. The principle of heat pump drying system is shown in Figure 3.

Currently, there exist three types of heat pump drying systems: closed, semi-open, and open,^[33] as depicted in Figure 4. The closed heat pump drying system is commonly utilized for drying Chinese medicinal herbs due to its operation at low temperature, typically below 50 °C. This low temperature ensured the preservation of heat-sensitive substances in the medicinal materials, albeit at the expense of drying rate and cycle duration. Addressing this limitation necessitated the continuous development of high-temperature heat pumps.^[35,36] Traditional heat pump drying technology not only extended processing times but also incurred excessive energy consumption. However, integrating innovative heat exchangers like phase change materials and heat pipes can bolster heat transfer efficiency within the drying medium, thereby accelerating moisture removal rates.^[37] The performance assessment of heat pump drying systems primarily hinged on metrics such as energy efficiency ratio, dehumidification energy consumption ratio, and drying thermal efficiency. It was observed that as the relative humidity of the air in the drying chamber decreases, the dehumidification amount, dehumidification energy consumption ratio and dehumidification heating performance coefficient of the system also decreased.^[38] Moreover, optimizing the flow rate and

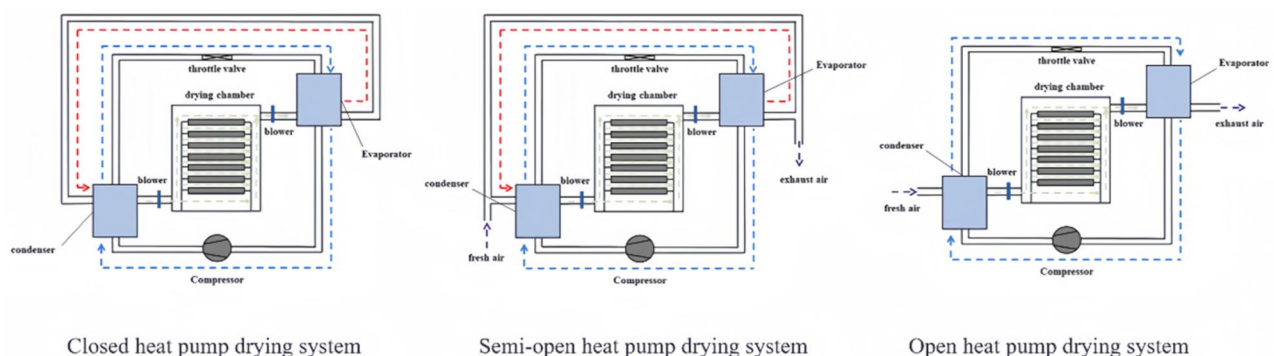


Figure 4. Three types of heat pump drying systems.^[34]

temperature of the circulating air can augment the drying efficacy of Chinese medicinal herbs. By elevating air flow velocity to sustain a consistent Reynolds number across varying vacuum pressures, it becomes feasible to counterbalance the reduction in heat transfer coefficient during the drying process due to decreased vacuum pressure, while simultaneously enhancing the mass transfer coefficient.^[39]

The heat pump drying technique not only boasted low energy consumption, rendering it an energy-efficient and environmentally friendly option, but also facilitated precise control over operational parameters. The implementation of a closed circulation system effectively prevented oxidation of Chinese medicinal herbs during the drying process. Nevertheless, its application was hindered by limitation on working temperatures. Looking ahead, the advancement of heat pump drying can be expedited through synergistic integration with other technologies, capitalizing on their individual strengths.

2.5. Microwave drying

In the process of microwave drying, a 915 MHz microwave field was employed to induce water molecules to oscillate at frequencies as high as 1.83 billion cycles per second.^[40] This rapid oscillation facilitated the absorption of microwave energy by the material, subsequently converting it into heat. As a result, moisture within the material evaporated, leading to an elevation in internal temperature relative to the surface temperature. This temperature differential established a gradient from the material's interior to its exterior. Figure 5 illustrated the experimental setup utilized for microwave drying.

Currently, microwave drying stands out for its remarkable ability to significantly expedite drying processes while ensuring the quality of materials. Its distinctive feature lies in its capability to align

moisture evaporation with heat transfer directions during drying, effectively preventing wrinkling of dried Chinese medicinal herbs.^[42] Furthermore, studies have revealed a correlation between pulse ratio and drying time, indicating that a larger pulse ratio correlates with shorter drying times and reduced energy consumption.^[43] Moreover, increasing microwave power leads to higher rehydration rates and stronger sterilization abilities during microwave drying.^[44,45] The electromagnetic field applied in microwave drying intensifies proportionally with microwave power, accelerating water evaporation and enhancing the drying rate.^[46,47] However, it's crucial to note that during the drying process, careful attention must be paid to ensure that microwave power density does not become excessively high. Otherwise, it may lead to internal overheating of Chinese medicinal herbs, causing charring and carbonization. Conversely, if the microwave power density is too low, the advantages of microwave drying may not be fully realized.^[48] In their study on microwave intermittent drying of *Rhizoma polygonati*, Wu et al.^[49] observed that higher power levels corresponded to shorter intermittent times, smaller loading capacities, and thinner slice thicknesses, resulting in faster drying rates. Similarly, Jin et al.^[50] found, through experiments on sweet potatoes with white hearts, that timely extraction of heat and water vapor in the drying oven accelerated water evaporation, thereby increasing the drying rate. Additionally, BEŞİR A et al.^[47] conducted microwave drying experiments on hawthorn slices, noting that lower microwave power levels correlated with longer heating times and higher rates of carotenoid degradation.

The microwave drying method has gained significant traction in the Chinese medicinal herbs drying industry owing to its rapid drying rate, environmental friendliness, and its capacity to maintain the quality, color, and shape of the medicinal materials. However, its utility is constrained when dealing with heat-sensitive Chinese medicinal herbs due to the elevated temperatures involved. Moreover, challenges in achieving uniform drying, limited penetration of medicinal components, high initial equipment investment costs, and complexities in operation and control hinder its wide-scale implementation in industrial settings. Furthermore, particular emphasis must be placed on mitigating radiation risks stemming from potential microwave leakage to safeguard personal safety. Addressing these limitations is crucial to fully harnessing the potential of microwave drying in the Chinese medicinal herbs industry.

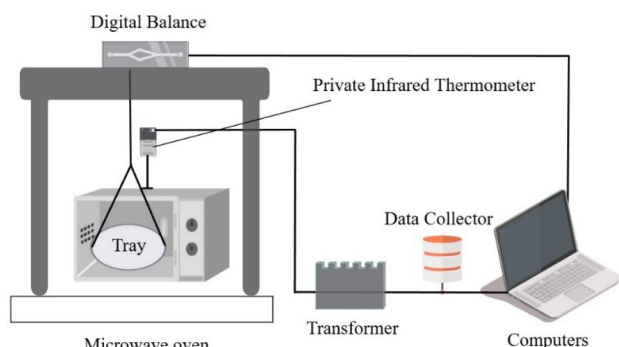


Figure 5. Microwave drying experimental equipment.^[41]

2.6. Far-infrared drying

Infrared radiation, situated between the wavelengths of visible light and microwaves, is categorized as an electromagnetic wave. The infrared spectrum ranged from 0.75 to 1000 μm , with distinctive molecular motion patterns corresponding to different infrared wavelengths. In contrast to hot air drying, far-infrared drying involved heat conduction from the interior to exterior of material. First, far infrared radiation with a wavelength greater than 4 nm is emitted from the heat source. When the electromagnetic wave reached the surface of the material, it induced changes in the internal molecular motion state. Upon penetrating the material, infrared rays cause internal molecules to vibrate at a frequency of 60,000 ~ 150,000 MHz, thereby generating heat through intermolecular friction and rapidly heating up the interior of the material. As a result of temperature differences, water is transported from inside to outside to complete the drying process.^[51]

Here, the material selectively absorbed radiation at specific wavelengths and efficiently transforms it into thermal energy. This process led to a gradual rise in temperature, facilitating water evaporation and enabling swift drying. However, the absorption process was selective. Only when the infrared emission frequency aligned with the natural frequency of the material molecules to be dried, did the internal molecules of the material resonate and absorbed the radiation energy.^[52,53] Figure 6 illustrates a schematic diagram of far-infrared drying.

Far-infrared drying technology has been shown to enhance the number of micro-holes on the surface and inside of materials, consequently improving the quality of dried products.^[55] The elevation of the far infrared drying temperature can supply additional energy and heat flux to the material, thereby

significantly enhancing both the heating rate and the evaporation rate of surface water. This results in an increased number and size of micropores within the material.^[56] Research by Li et al.^[57] revealed that *Angelica sinensis* slices subjected to far-infrared drying exhibited a uniform arrangement of pores between cells. Additionally, due to the consistent heating internally and externally, the dried *Angelica sinensis* displayed minimal wrinkling or deformation, resulting in an overall improved product quality. In a study by Zhou et al.,^[58] it was found that the thickness of *Huai yam* significantly influenced the thermal effect of far-infrared radiation. Specifically, thinner slices experienced a stronger thermal effect, leading to faster drying rates and shorter drying times. Therefore, careful consideration of material slice thickness is paramount when employing far-infrared drying, as the energy of far-infrared radiation diminishes gradually with increasing depth of penetration.

Far-infrared drying, when coupled with ultrasonic pretreatment, significantly enhances the quality of Chinese medicinal herbs. Zhang et al.^[59] observed that employing ultrasonic-assisted drying techniques increases the quantity of micropores on the surface of Chinese wolfberry, mitigate damage to epidermal cells, and reduces mass transfer resistance, thereby accelerating the drying process. Pei et al.^[54] noted that longer ultrasonic pretreatment times correlated with reduced total drying times, higher levels of main volatile components, and increased contents of crocin and flavonoids. Furthermore, Zhang et al.^[60] demonstrated that appropriate ultrasonic pretreatment slightly increased energy consumption but substantially improved far-infrared drying efficiency. This enhancement was more pronounced when increasing ultrasonic power compared to extending pretreatment time. Shang et al.^[61] found that ultrasonic pretreatment, combined with far-infrared drying, significantly altered the surface structure of licorice slices, creating more microporous channels, this restructuring facilitated mass and heat transfer during drying, resulting in reduced drying times and moisture content.

The far infrared drying process presented notable advantages such as convenient operation and cost-effectiveness, making it well-suited for industrial-scale production. Nonetheless, due to the restricted penetration capability of far infrared radiation, precise control over slice thickness was crucial. Moving forward, it is imperative to delve into more advanced techniques in combined far infrared drying technology. Such as far infrared assisted refractance window drying,^[62] far-infrared radiation heating assisted pulsed vacuum

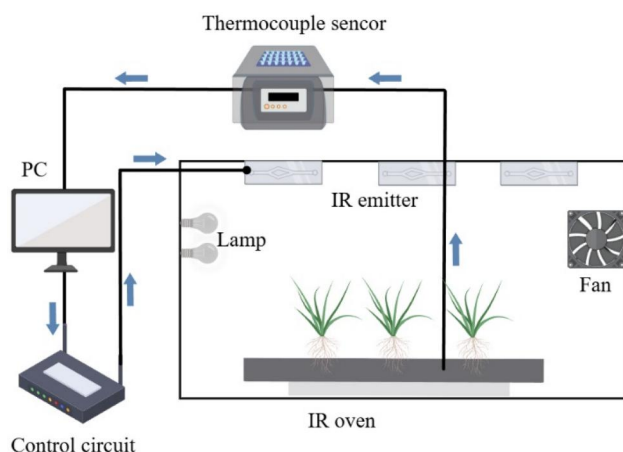


Figure 6. Schematic diagram of far infrared drying.^[54]

drying,^[63] far infrared-assisted freeze-drying^[64] and so on. By doing so, we can fully unlock its potential for heightened efficiency and energy conservation, thereby driving forward the development of the Chinese medicinal herbs drying industry. This strategic focus on innovation promises to pave the way for sustainable and impactful advancements in the field.

2.7. Vacuum drying

The vacuum drying process, also referred to as decompression analytical drying, entails exposing the material to a vacuum and negative pressure environment. Vacuum drying is primarily achieved through the following three coordinated processes: (1) The heat source transferred heat to the material, raising its temperature and vaporizing the wet component; (2) The liquid state of the wet part inside the material migrated to its surface in order to facilitate vaporization, thus constituting a mass transfer process for the internal wet component. (3) The vaporized wet component steam migrated to the drying chamber space, causing it to break off from the surface of the material and exit the drying chamber.^[65] By carefully raising the temperature, the material's melting point can be attained under negative pressure conditions, thus enabling efficient drying.^[66] The vacuum drying system typically comprises a vacuum dryer, a vacuum system, a heating system, and a control system, as illustrated in Figure 7.

Yan et al.^[68] identified two distinct stages in the vacuum drying process of *Cornus officinalis*: a significant constant-speed drying stage followed by a slow-down drying stage. During the constant-speed drying

stage, free moisture within the material is rapidly removed, while surface moisture continues to evaporate. Consequently, the drying temperature plays a crucial role in this stage. In contrast, during the deceleration drying stage, the removal of bound moisture occurs, and although surface moisture evaporates swiftly, the transfer of internal moisture to the surface is relatively slow. Therefore, the drying temperature has a relatively minor impact during this stage. Liu et al.^[69] investigated the characteristics of hawthorn under various vacuum conditions. They found that when the thickness of hawthorn was 0.15 cm, the vacuum degree had no significant effect on the drying rate. However, as the thickness increased to 0.30 and 0.50 cm, the drying rate rose with the increase in vacuum degree. This phenomenon occurred because lower pressure reduced the boiling point of water, facilitating easier evaporation. Nonetheless, it also affected heat transfer, resulting in minimal influence on the average drying rate.^[70]

Pulsed vacuum drying is an emerging technology that operates by subjecting the drying chamber to alternating cycles of vacuum and atmospheric pressure. This pulsation induces a tunnel effect within the material, resulting in the formation of numerous pore structures internally and thereby accelerating the drying rate. Lower pressures correspond to lower boiling points for water, facilitating easier evaporation, albeit with minimal impact on the overall average drying rate.^[71] Zhang et al.^[72] investigated the vacuum pulsation drying characteristics of *Poria cocos* and observed if the vacuum time was too short, the drying rate decreased, and time consumption increased. Conversely, if the vacuum time was too long, the partial pressure of water vapor on the surface of *Poria*

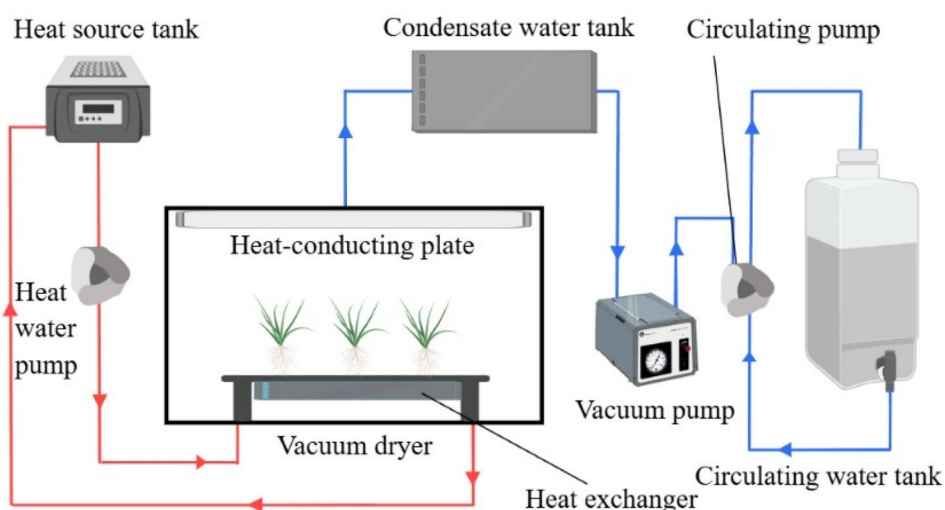


Figure 7. Schematic diagram of vacuum drying.^[67]

cocos tended to balance with that in the drying chamber, leading to a reduction in the dehydration rate of materials. Similarly, if the duration of atmospheric pressure was too short, the drying rate became equivalent to constant vacuum drying. On the other hand, if the duration of atmospheric pressure was too long, it shortened the vacuum drying time for *Poria cocos*, which is also unfavorable for improving the drying rate. Therefore, maximizing the advantages of vacuum pulsation drying requires setting reasonable durations for both vacuum and atmospheric pressure cycles. This ensures optimal drying efficiency and performance.

The vacuum drying process provide several advantages, such as the ability to operate at low temperatures and achieve rapid drying rates. Moreover, the utilization of a relatively hypoxic environment during drying effectively inhibited the oxidation of medicinal components and prevented pigments from undergoing browning. This feature was particularly beneficial for preserving the quality and potency of medicinal herbs during the drying process.

2.8. Vacuum freeze drying

Compared to classical freeze drying, vacuum freeze drying was a more advanced method. While both processes utilize low temperatures to extract water from a substance, vacuum freeze drying created a low-pressure environment that facilitates the evaporation of water from the substance, resulting in a faster drying process. The vacuum freeze-drying process

could be delineated into five distinct stages: pre-freezing, refrigeration for water capture, vacuum extraction, drying, and heat preservation. Moisture removal from frozen materials occurs *via* sublimation, entailing the direct conversion of solid to gas, to accomplish drying. The schematic diagram depicting vacuum freeze drying is presented in Figure 8.

The first step in drying high-quality Chinese medicinal herbs was to establish suitable pre-freezing temperature, sublimation temperature, vacuum level, and partition temperature. Lower pre-freezing temperatures resulted in higher freezing rates, and an increased number of crystal nuclei formed in the material enhanced the quality of the freeze-dried product. However, this process was time-consuming and inefficient.^[74] The adjustment of sublimation temperatures, whether too high or too low, can induce variations in the morphological parameters of cells. Insufficient sublimation at lower temperatures during the initial stage might result in incomplete removal of ice crystals, which subsequently melted during the analytical drying phase, leading to material softening and deformation. Conversely, higher sublimation temperatures induced rapid sublimation of ice crystals on the cell surface, while the internal ice crystals underwent slower sublimation, resulting in material hardening and wrinkling.^[75] The drying rate increased with the elevation of vacuum degree but eventually decreased after reaching a certain threshold. Therefore, determining the optimal vacuum degree through experimental investigations for different materials is essential. As the partition temperature

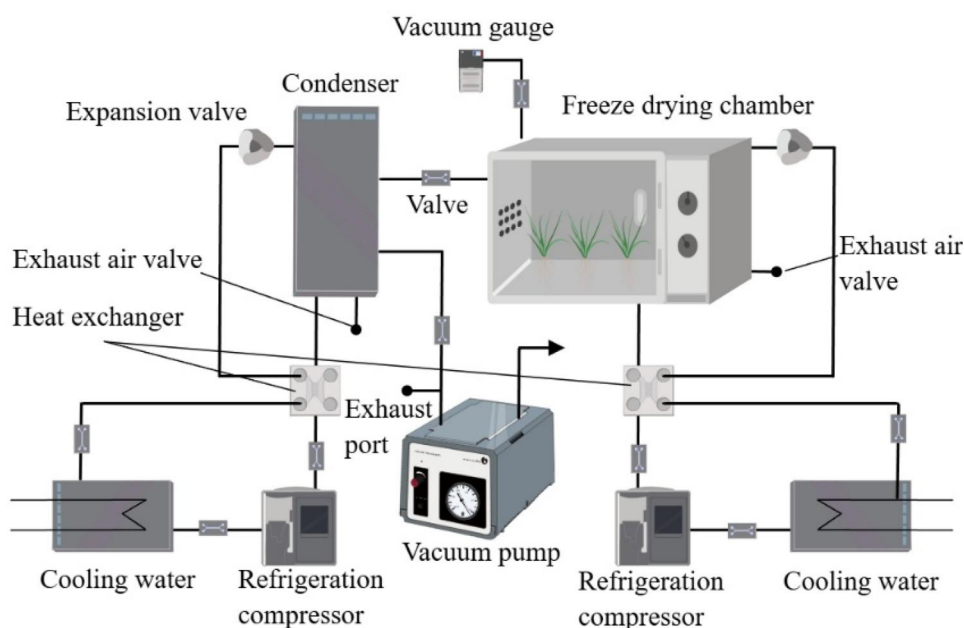


Figure 8. Schematic diagram of vacuum freeze drying.^[73]

increased, the heat transfer driving force augmented, thereby facilitating moisture evaporation. However, excessive temperatures can lead to uneven heating and wrinkling of the material.^[74]

Furthermore, the eutectic point and the melting point of the material are also crucial parameters in vacuum freeze drying. Yang et al.^[76] found that achieving complete freezing of *Gastrodia elata* during vacuum freeze-drying experiments necessitated a pre-freezing temperature below the eutectic point of 5–10 °C. It was important to ensure that the drying temperature in the sublimation process does not exceed its eutectic point to prevent material wrinkling or expansion.

In general, vacuum freeze drying stood out as one of the most extensively researched novel drying technologies. This method, conducted in a low-temperature environment, offers numerous advantages. The use of a sealed drying chamber effectively prevents the oxidation of Chinese medicinal herbs. Water sublimation preserved the original pre-frozen water space, preventing wrinkling of medicinal materials and cell damage while effectively maintaining their appearance quality, color, and shape. Additionally, this process retained the bioactive components present in Chinese medicinal herbs. However, the primary drawback of vacuum freeze drying lay in its exorbitant equipment costs, prolonged drying duration, and substantial energy consumption. The sublimation drying process alone accounted for 48% of the overall energy expenditure associated with freeze drying.^[77] Simultaneously, diverse medicinal materials required distinct process design and control measures at different stages, directly impacting the performance of freeze-dried products and associated equipment.

2.9. Radio frequency drying

The radio frequency (RF) spectrum represented electromagnetic waves operating within the frequency range of 1 to 300 MHz. To address industrial, scientific, and medical needs, the Federal Communications Commission (FCC) has delineated three specific frequency bands: 13.56, 27.12, and 40.68 MHz. When subjected to an alternating electric field, materials experience dielectric heating, resulting in two distinct phenomena. Firstly, there is the migration of positively charged ions in one direction and negatively charged ions in the opposite direction. Secondly, dipoles underwent continuous rotation due to the high-frequency variation of the magnetic field's orientation. Throughout this process, friction among

molecules converted electromagnetic energy into heat, thereby causing the temperature of treated materials to increase.^[78–80] The schematic diagram illustrating radio frequency drying is presented in Figure 9.

The RF drying system is typically classified into two distinct types: the 50-Ω self-matching system and the free-running oscillator RF heating systems, as illustrated in Figure 10. The free-running oscillator system is widely utilized due to its simplicity, flexibility and cost-effectiveness compared to the 50-Ω system, making it a preferred choice in the drying industry.^[82] Li et al.^[83] conducted an RF drying

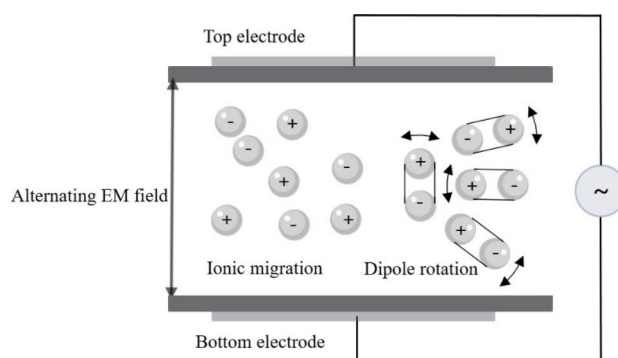


Figure 9. Schematic diagram of radio frequency drying.^[78]

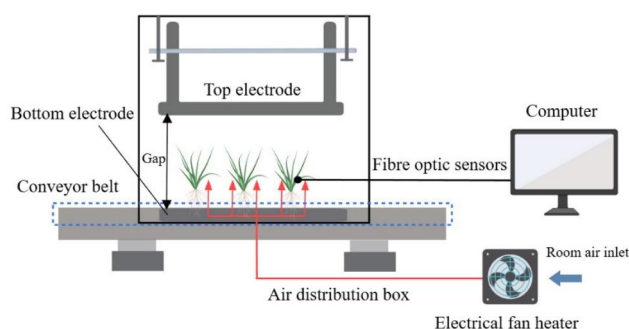
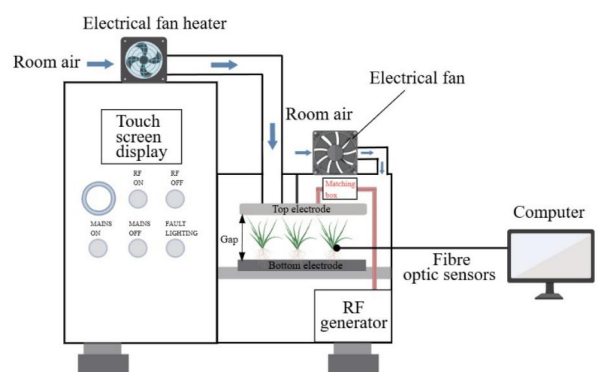


Figure 10. Diagram of 50-Ω self-matching and free-running oscillator radio frequency heating systems.^[81]

experiment on figures and observed that increasing the plate spacing resulted in a prolonged duration for the figures to reach a temperature of 90 °C through RF heating. This is attributed to the widening spacing between the electrodes, which weakens in the electric field strength, thus reducing the molecular motion rate and extending the drying time. Consequently, an increased amount of phenolic compounds is generated during the drying process. It is essential to carefully regulate the plate spacing during the drying process to prevent material coking caused by excessively small plate spacing and high electric field strength. Conversely, excessive plate spacing can prolong the drying time and potentially lead to enzymatic browning.

The RF drying process, akin to microwave drying, facilitates thorough penetration into the material's interior. In Comparison with microwave drying, RF drying boasted superior penetration capabilities, enhanced drying efficiency, reduced energy consumption, and the ability to inhibit enzyme activity, among other advantages. The moderate temperatures achieved through radio frequency drying help alleviate the risk of medicinal material cracking induced by high temperatures, thereby minimizing the loss of volatile medicinal components. In the realm of RF drying, it is crucial to prevent excessive concentration of RF energy on specific areas, as this can lead to overheating and subsequent edge coking. Consequently, there is a need for further exploration into the alteration patterns of dielectric properties during RF drying.

2.10. High voltage electric field drying

High voltage electric field drying technology, also referred to as Electrohydrodynamic drying technology (EHD) or Corona wind drying technology, represented a novel form of non-thermal drying technology. The entire system of high voltage electrostatic field comprised two parallel electrode plates. The electric field intensity could be adjusted by altering the output voltage or modifying the distance between the electrodes. The equilibrium of positive and negative charges ensured equality of the electric field between the two plates. A coronal wind was generated between the emitter and the collector, which subsequently impacted the material surface, facilitating heat and mass transfer. This process promoted the detachment of water molecules from the material, ultimately achieving a drying effect.^[84–86] The experimental setup for high voltage electric field drying is depicted

in Figure 11, while a schematic illustration of EHD airflow generation process for positive corona discharge is presented in Figure 12.

Currently, high voltage electric field drying has found application in the dehydration of Chinese medicinal herbs such as *Chinese wolfberry*,^[89] *yam*^[90] and *Gastrodia elata*,^[91] resulting in enhanced drying rate and improved quality of the medicinal products. Taghian et al.^[92] investigated the effect of decreasing the voltage and increasing the electrode gap on the internal temperature of mushroom slices, noting a reduction in the time required to achieve a constant moisture content. Anukiruthika et al.^[93] observed that the total number of needles in the emitter influences the electric field strength. Increasing the number of needles enhances the electric field strength and expands the ionization area, ultimately leading to an increase in discharge current. Dalvi-Isfahan et al.^[84] speculated that the high electric field intensity could alter the structure of polar molecules, facilitating chemical reactions by lowering the activation energy. Enhanced electric fields can also induce electrical perforation and penetration of the cell membrane. Ni et al.^[94] investigated the drying characteristics of *Chinese wolfberry* under a high voltage electric field and found that increasing needle spacing decreased the effective diffusion coefficient of water within *Chinese wolfberry* and the rate of cell damage. Sriariyakul et al.^[95] developed a drying system combining hot air and fa-infrared with high voltage electric field. Their results indicated that the rate of moisture reduction increased with the increased of dry air, controlled surface temperatures, and electric field intensity. In addition to drying, high voltage electric fields find applications in food freezing, refrigeration, thawing, and extraction of bio-compounds.^[93]

In contrast to conventional energy-intensive drying techniques, high voltage electric field drying demonstrated reduced energy consumption and accelerated drying rates. Furthermore, it represented a novel

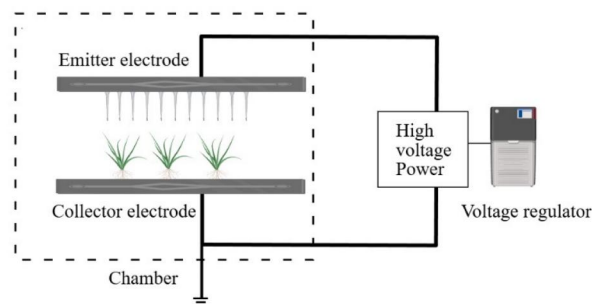


Figure 11. Diagram of high voltage electric field drying.^[87]

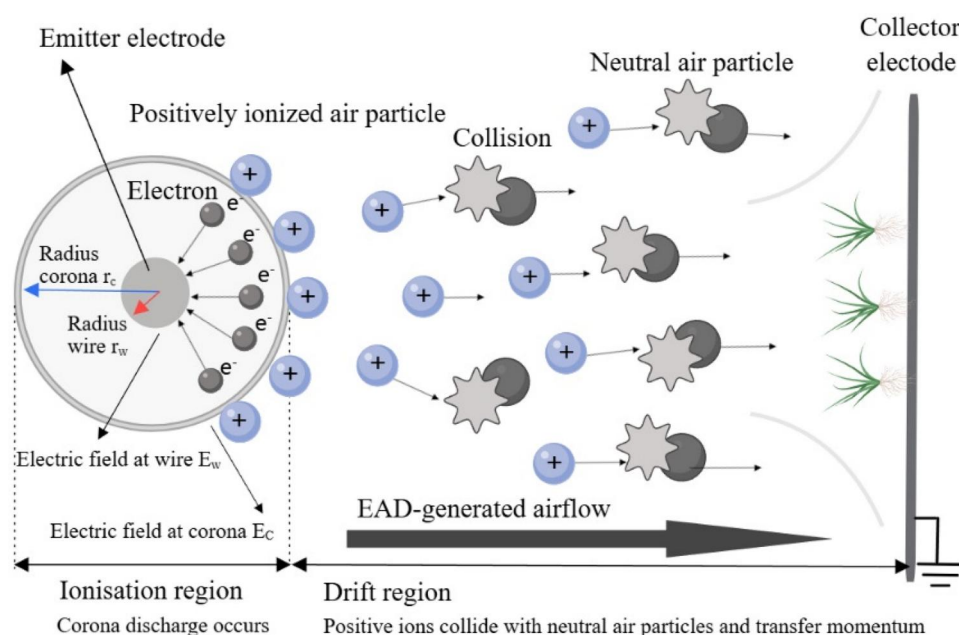


Figure 12. Schematic illustration of EHD airflow generation process for positive corona discharge.^[88]

Table 1. Comparison of various single drying technologies.

Drying technology	Investment cost	Drying time	Temperature range	Maintenance	Energy consumption	References
Shade drying	–	Long	Room temperature	–	0	[96]
Hot air drying	Lower	Long	50 °C~55 °C	Easy	4.167 kWh/kg	[97]
			60 °C		334.8 MJ/kg	[98]
Solar drying	Lower	Long	34.5 °C~57.6 °C	Easy	4.509 kWh/kg	[99]
			41.8 °C~61.2 °C		11.29 kWh/kg	[100]
Heat pump drying	Low	Long	45 °C~55 °C	Average	0.357 kWh/kg ~0.380 kWh/kg	[38]
			50 °C~55 °C		0.813 kWh/kg	[97]
Microwave drying	High	Short	60 °C~100 °C	Difficult	1.98 kWh/kg~3.72 kWh/kg	[101]
			105 °C		5.81 MJ/kg	[102]
Far-infrared drying	Low	Short	45 °C~65 °C	Difficult	2.86 kWh/kg~7.31 kWh/kg	[103]
			35 °C~55 °C		2.40 kWh ~3.15 kWh	[104]
Vacuum drying	High	Long	81 °C	Average	4.43kJh/kg	[72]
			50 °C~70 °C		9.34 kW/kg	[105]
Vacuum freeze drying	Higher	Long	25 °C~65 °C	Difficult	15.617 kJ/g	[106]
			37 °C		4.1 kJ/g	[107]
Radio frequency drying	High	Short	–	Average	3.38 kWh/kg	[108]
			65 °C		4241.4389 kJ/kg	[109]
High voltage electric field drying	Low	Short	65 °C	Easy	17.25 ± 0.38 MJ/kg	[110]

non-thermal technology characterized by minimal thermal energy loss and significant energy-saving benefits.^[87] A major advantage of high voltage electric field drying was the concurrent sterilization during the drying process. Since there are no mechanical components involved, equipment costs are minimized, operation is simplified, and achieving automatic control becomes effortless. Consequently, maintenance requirements are minimal.

The equipment and technologies discussed above are currently employed in the drying process of Chinese medicinal herbs. A comparison of the drying characteristics for each individual technology is presented in Table 1.

2.11. Other combined drying technologies

Typically, a singular drying method struggled to attain the desired effect, as each drying technique possesses inherent limitations. Nonetheless, through the amalgamation of these methods, such constraints could be overcome, complementary advantages could be unlocked. By integrating two or even three technologies, the quality of numerous Chinese medicinal herbs can witness significantly enhancement, accompanied by a notable retention of pharmaceutical active ingredients, alongside reduced drying durations and energy conservation. Consequently, the fusion of drying technologies emerged as a domain ripe with potential for advancement. Table 2 provided a comprehensive

Table 2. Correlation studies of different combined drying.

Combined drying technology	Herbs	Drying effect	References
Microwave – far infrared drying	<i>Ginseng</i>	Compared to single far-infrared drying, combined drying exhibits enhanced efficiency in terms of drying speed, ginsenoside content value, and surface shrinkage rate reduction.	[111]
Microwave – vacuum freeze drying	<i>Passion fruit peel</i>	The rehydration ratio, total phenol content, and antioxidant activity of the dried pericarp of passion fruit were found to be significantly higher.	[112]
Microwave – hot air drying	<i>Scutellaria baicalensis</i>	The baicalin content in the combined drying method exhibited higher levels compared to those observed in single microwave and hot air drying techniques.	[113]
	<i>Ginger</i>	Combined drying accelerates the drying process, and when the microwave power was increased from 0.6 W/g to 0.9 W/g, the microstructure was more damaged, the release of bioactive compounds was improved, and the antioxidant activity was also increased.	[114]
Microwave – vacuum drying	<i>Astragalus membranaceus</i>	The shortened drying time and vacuum environment effectively lower the material's drying temperature, thereby mitigating thermal damage caused by high temperatures and enhancing overall quality.	[115]
	<i>Okra</i>	The total phenol and flavonoid contents in MVD products were not significantly different from vacuum freeze drying, but significantly higher than HAD. MVD was an efficient and energy-saving technology that retains color, flavor, nutrients, and bioactive compounds in the product.	[116]
	<i>Ganoderma lucidum</i>	The quality of <i>ganoderma lucidum</i> concentrated extract obtained by the combined drying method is close to that of freeze-dried extract, and obviously better than that of conventional vacuum drying extract.	[117]
Pulse spouted microwave-vacuum drying	<i>Okra</i>	The samples dried by PSMVD showed higher quality and more uniformity compared to MVD.	[118]
Pulse spouted microwave-freeze drying	<i>Yam</i>	Considering the high efficiency and comparable quality compared with FD evaluated by principal component analysis (PCA), PSMFD was a promising drying technique.	[119]
Solar – heat pump drying	<i>Chinese wolfberry</i>	Under the same drying temperature setting, the combined drying of solar energy and heat pump saved 28.8% of electricity compared with the drying of heat pump alone.	[120]
	<i>Misai Kucing</i>	Kucing's leaves, flowers, stems, and mix along with the color characteristics, solar-assisted heat pump-dried samples afforded better results compared to solar drying.	[121]
Heat pump – high voltage electric field drying	<i>Yam</i>	The unit energy consumption of combined drying can be reduced by 6%~19% compared to single heat pump drying at the same temperature.	[90]
Heat pump – hot air drying	<i>Gynuraproculumbens</i> Leaves	The <i>Gynuraproculumbens</i> Leaves obtained by combined drying have low water content and green color, which not only shortens the drying time, but also significantly improves the quality of medicinal materials.	[122]
Heat pump – microwave drying	<i>Stichopus japonicus</i>	Compared with the single heat pump drying, the drying rate was significantly increased, and the rehydration of dried sea cucumber was better under the same temperature.	[123]
Radio frequency – hot air drying	<i>Juglans regia</i> L.	The indexes of <i>juglans regia</i> L. dried by radio frequency and hot air are within the industrial production range, and the quality of <i>Juglans regia</i> L. dried by single hot air is better.	[124]
Far infrared drying – hot air drying	<i>Alfalfa</i>	Compared with single drying, combined drying has higher nutrient retention and lower energy consumption.	[125]
	<i>Panax Notoginseng</i> roots	IR-HAD is a promising drying technology, especially at 50 °C, as it allowed high retention of PNS (about 12.5%) and low unit energy consumption (about 19.5 kWh/kg).	[126]
Far infrared drying – heat pump drying	<i>Dimocarpus longan</i> Lour.	Maintains good color and smell, and preserves the amino acid content to a greater extent.	[127]
Far infrared drying – pulsed vacuum drying	<i>Chinese wolfberry</i>	Prevent the degradation of vitamin C in Chinese wolfberry, and retain the content of flavonoids and betaine to a greater extent.	[128]
	<i>Codonopsis pilosula</i>	Compared with hot air drying, the total phenolic content and antioxidant activity of <i>C. pilosula</i> by using ultrasonic synergy vacuum far-infrared drying were increased by 29.47% and 8.67%, respectively, and a reduction in color contrast of 16.19%.	[129]
Infrared – freeze drying	<i>Turmeric</i>	Compared with freeze drying, infrared freeze drying shortened the drying time by 21.27% and significantly reduced the energy consumption by 25.19% ($p < 0.05$).	[130]
Vacuum freeze – hot air drying	<i>Hippophae rhamnoides</i> L.	Combined drying not only maintains the color of fruits better, but also makes the content of VC, total phenols and flavonoids higher.	[131]
Microwave vacuum – hot air drying	<i>Panax notoginseng</i>	Compared with single hot air drying, combined drying can shorten the drying time by 44.6%~60.0%, and the content of total saponins in dried <i>Panax notoginseng</i> products is up to 10.2%.	[132]
Vacuum freeze – vacuum microwave	<i>Burdock</i>	Compared with vacuum freezing drying, the drying time was shortened by 53.9%, and the energy consumption was reduced by 57.9%. The total phenol content in the products of vacuum freezing combined with vacuum microwave drying was not significantly different from that of vacuum freezing drying.	[133]

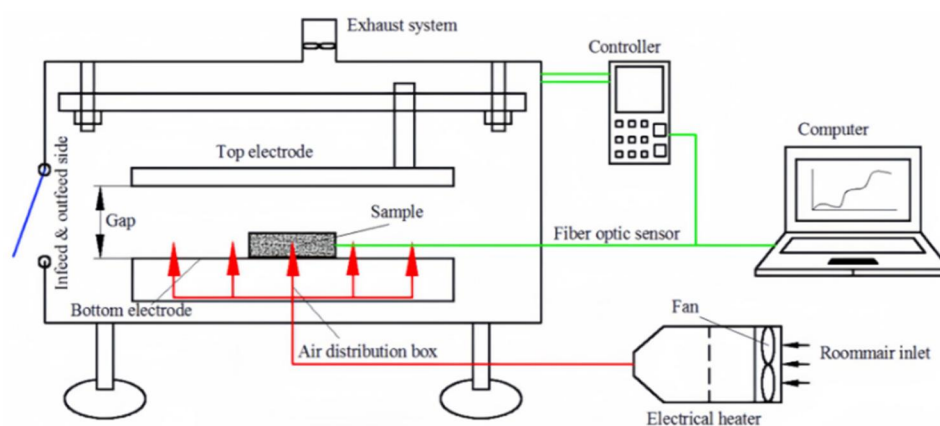


Figure 13. Diagram of the RF/hot air drying system.^[134]

overview of the drying efficiencies attained through diverse combined drying techniques.

In the process of drying, it is essential to select the most suitable combined drying method based on the characteristics of the material. This requires consideration of the composition, form, and water content of the material, as well as a comprehensive assessment of production processes and equipment conditions. For materials containing heat-sensitive ingredients, vacuum freeze-drying is necessary. However, due to its time-consuming nature and high energy consumption, microwave^[133] and hot air^[131] can be used in combination to significantly improve efficiency. Low temperature heat pump drying effectively preserves the sensory quality of medicinal materials but suffers from low efficiency. Therefore, it can be combined with high pressure electric field,^[90] hot air,^[122] microwave^[123] or other drying methods. Furthermore, for easily oxidized materials, vacuum drying is essential. Microwave drying can also be utilized to expedite the overall drying process.^[115]

Song et al.^[134] introduced a novel hot air-assisted radio frequency drying technology for drying *red dates*, with the equipment principle illustrated in Figure 13. Experimental results indicate that the combined drying method significantly surpasses conventional hot air drying techniques in terms of color retention, vitamin C (V_C) content, and total flavonoid content. Xu et al.^[135] devised a combined drying technology incorporating infrared and hot air, as showed in Figure 14, and evaluated its effects on *chrysanthemum* quality compared to single drying methods. The findings revealed that *chrysanthemums* subjected to combined drying exhibited higher retention rates of total phenols, superior color preservation, and more uniform cell distribution compared to those dried solely using hot air.

Compared to single drying, the combined drying technology not only enhances the preservation of medicinal ingredients and sensory quality but also facilitates flexible adjustment of drying parameters based on specific requirements, thereby further reducing energy consumption. Comparative analysis of energy consumption among different drying methods is presented in Tables 3 and 4. Meng et al.^[90] investigated the joint drying characteristics of heat pump-high voltage electric field for yam slices. It was observed that the moisture evaporation of yam slices in joint drying was faster than that in single heat pump drying, with a notable increase in voltage resulting in an accelerated moisture evaporation process. At the same temperature, the drying time required by combined drying is shortened by 14–22% compared with that of single heat pump drying, and the energy consumption per unit can be reduced by 6–19%. Yan et al.^[120] examined the drying characteristics of *Chinese wolfberry* in a solar heat pump combined system and found that the combined drying method led to a 28.8% reduction in electricity consumption compared to single heat pump drying. When compared to traditional drying methods, the cost savings for drying one ton of *Chinese wolfberry* amounted to approximately 448 RMB. Furthermore, it also resulted in a conservation of 121.14 kg of standard coal and a reduction in CO₂ emissions by 317.39 kg. Zhang et al.^[133] employed a combined drying method to dry burdock slices, and the findings demonstrated that when compared to single hot air drying, the integration of hot air with vacuum microwave drying reduced the drying time by 90.0% and decreased energy consumption by 80.1%. Furthermore, in comparison to single vacuum freezing drying, combining vacuum freezing with vacuum microwave drying resulted in a reduction of the drying time by 53.9% and energy consumption by 57.9%.

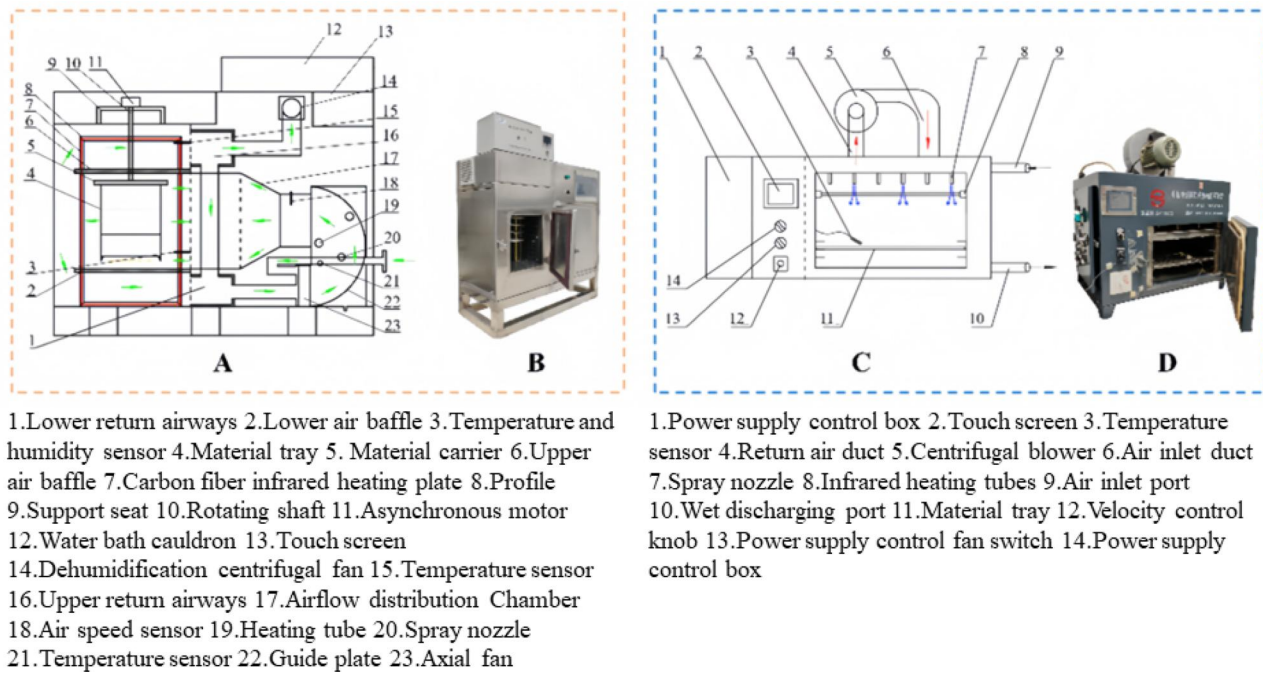


Figure 14. The schematic and physical diagram of dryers (the arrows represent the direction of the airflow).^[135]

Table 3. Energy consumption of different drying methods for drying *Eucommia ulmoides* male flowers.^[136]

Drying method	Drying time/h	Drying energy consumption/kJ·g ⁻¹
Microwave-vacuum freeze drying	6.0 ± 0.08	51.65 ± 0.66
Vacuum freeze drying	14.1 ± 0.32	213.27 ± 4.85
Hot air drying	7.0 ± 0.08	86.80 ± 0.95
Heat pump drying	7.5 ± 0.13	51.74 ± 0.87
Far infrared drying	7.5 ± 0.08	97.02 ± 0.99

Table 4. Effect of drying methods on the drying time, drying energy consumption of burdock root slices.^[133]

Drying method	Drying time/h	Drying energy consumption / (KW h) kg ⁻¹
Hot air drying	8.30 ± 0.35	84.40 ± 0.58
Vacuum microwave drying	0.48 ± 0.03	10.30 ± 0.32
Vacuum freeze drying	16.30 ± 0.50	102.80 ± 0.80
Hot air-vacuum microwave drying	0.83 ± 0.09	16.80 ± 0.60
Vacuum freezing-vacuum microwave drying	7.52 ± 0.18	43.28 ± 0.90

Wang et al.^[137] analyzed the characteristics and energy consumption of microwave-hot air coupled drying of peanut, and the results showed that the total energy consumption of combined drying was reduced by 37.1% and 8.6%, respectively, compared with low-intensity microwave and high-intensity microwave drying, and better quality peanut could be obtained. Li et al.^[138] utilized a combination of hot air and microwave to dry 300 g of *Phyllanthus emblica* L. resulting in a combined drying time of only 8.4125 h, which led to an energy savings of 14.08% compared to single hot air drying. Wang et al.^[139] investigated various drying methods for okra, including

far-infrared drying, hot air drying, vacuum freeze drying, vacuum microwave drying, vacuum freeze combined with vacuum microwave drying, and hot air combined with vacuum microwave drying. The findings revealed that the energy consumption of vacuum freeze combined with vacuum microwave drying was approximately one-third of that observed in vacuum freeze drying. Similarly, the energy consumption of hot air combined with vacuum microwave drying was about one-fifth of that seen in hot air drying.

The combined drying technology presented a plethora of advantages, encompassing superior material quality, accelerated drying rates, and decreased energy

usage. As a result, the transition from single drying methods to combined drying techniques emerged as a prominent trend in future development. This evolution bore considerable application significance and heralded promising prospects for further advancement.

3. Influence of drying technology on Chinese medicinal herbs bioactive compounds

The drying process plays a pivotal role in the processing of Chinese medicinal herbs, where the choice of drying methods and parameters profoundly impacts the quality of these herbs. Beyond evaluating the quality of medicinal herbs through criteria like moisture content, color, taste, and morphology, it becomes imperative to scrutinize the presence of bioactive constituents within these materials, encompassing polysaccharides, total phenols, flavonoids, saponins, volatile oils, and more.

3.1. Polysaccharide content

Polysaccharides are crucial active constituents found in a multitude of Chinese medicinal herbs. Their biological efficacy hinges not only on factors like molecular weight and monosaccharide composition but also on intricate structural features. These polysaccharides manifest diverse functionalities, including immune regulation, anti-tumor activity, blood sugar reduction, and anti-aging effects. Hence, ensuring the preservation of polysaccharides throughout the drying process assumes crucial importance.

Table 5 compiles studies investigating the polysaccharide content in Chinese medicinal herbs. It is crucial to note that the thermal conditions utilized during processing may induce alterations in the physical and chemical properties of polysaccharides, potentially leading to protein denaturation and subsequent reduction in antioxidant activities.^[145] Additionally, polysaccharides are susceptible to decompose and deteriorate at high temperature, resulting in a decrease in their content.^[148] Various drying methods exert differing effects on the physical and chemical properties as well as antioxidant activities of polysaccharides. To extract polysaccharide as the target product, it is imperative to avoid employing high-temperature drying methods such as microwave drying and hot air drying. low-temperature drying process are highly effective in inhibiting enzyme degradation and preserving a significant amount of polysaccharides in Chinese medicinal herbs.

Some Chinese medicinal herbs are susceptible to enzymatic browning or oxidation reactions during the drying process, resulting in tissue browning or fading. Typically, color protection pretreatment is conducted before drying to mitigate these effects. This pretreatment not only helps prevent color fading of the medicinal materials to some certain extent but also has been found to significantly enhance the content of polysaccharides.

Dai et al.^[149] conducted a study on processed dried *Dendrobium officinale* flowers using color protection and hot air drying technology. Their research found

Table 5. Research on polysaccharide contents in traditional Chinese medicinal herbs.

Herbs	Drying method	Polysaccharide content(minimum)	Corresponding drying method	Polysaccharide content(maximum)	Corresponding drying method	References
<i>Dendrobium officinale</i>	Hot air drying (60 °C, 90 °C), vacuum drying (60 °C, 90 °C), vacuum freeze drying (-40 °C)	83.45%	90 °C vacuum drying	89.99%	Vacuum freeze drying	[140]
<i>Abelmoschus esculentus</i>	Shade drying, hot air drying, far infrared drying, freeze drying, microwave vacuum drying	67.0%	Shade drying	69.6%	Vacuum freeze drying	[141]
<i>Cornus officinalis</i>	Hot air drying, vacuum drying, vacuum freeze drying	85.91%	Vacuum drying	86.77%	Vacuum freeze drying	[142]
<i>Yam</i>	Hot air drying, vacuum drying, microwave drying, vacuum freeze drying	8.33%	Hot air drying	18.07%	Vacuum freeze drying	[143]
<i>Burdock</i>	Shade drying, hot air drying, far infrared drying, freeze drying, microwave vacuum drying	43.82%	Hot air drying	54.61%	Freeze drying	[144]
<i>Ligusticum chuanxiong</i>	Hot air drying, vacuum drying, freeze drying	81.51%	Hot air drying	87.72%	Freeze drying	[145]
<i>Jujube</i>	Hot air drying, freeze drying, far infrared drying, radio frequency drying	66.19%	Hot air drying	76.06%	Freeze drying	[146]
<i>Symphytum officinale</i>	Hot air drying, vacuum drying, freeze drying	59.03%	Hot air drying	74.82%	Freeze drying	[147]

that a significant increase in polysaccharides content after color protection treatment. However, they also noted the importance of preventing non-enzymatic browning caused by excessive concentration of color protection agent. Xiao et al.^[150] investigated the processing of iron stick yam slices with hot air drying following six different single or compound color protection treatments. Results indicated that the polysaccharide content in the untreated group was the lowest, probably due to improved polysaccharide extraction rates following a certain degree of heat treatment. Qiu et al.^[151] implemented color protection treatment before hot air drying of longan. Their study demonstrated that pre-drying color protection led to reduced polysaccharide loss, with polysaccharide content increasing as the duration of color protection pretreatment extended. This could be attributed to the presence of color protection agents hindering Maillard reaction between sugars and other compounds during the drying process.

3.2. Total phenolic content

Table 6 compiles studies investigating the polyphenol content in Chinese medicinal herbs. Phenolic

compounds are categorized as bioactive substances and are known to be sensitive to heat. The structural integrity of phenolic compounds may deteriorate under aerobic conditions, elevated temperatures, and exposure to light, leading to the degradation of these valuable constituents. When preserving a significant quantity of polyphenols is the objective, low-temperature and vacuum drying are recommended to effectively prevent the oxidation of polyphenols.

Meng et al.^[152] conducted a study examining the effects of various drying methods, including solar drying, vacuum drying, microwave drying, frying, vacuum freeze drying, hot air drying and other on the quality of nettle leaf. Their findings revealed that appropriate heating promotes the non-enzymatic transformation of phenolic precursors into phenolic aldehyde molecules, thus facilitating the generation of phenolic substances. As a result, vacuum drying yielded nettle leaf with the highest total phenol content and the least oxidation loss of phenolic substances. Similarly, Gao et al.^[153] explored the impact of different drying techniques on the overall phenolic content of dandelion leaves, finding that vacuum drying resulted in the highest total phenolic content. The decomposition of phenolic substances is accelerated in

Table 6. Research on polyphenol contents in Chinese medicinal herbs.

Herbs	Drying method	Total phenolic content (minimum)	Corresponding drying method	Total phenolic content (maximum)	Corresponding drying method	References
Nettle leaf	Solar drying, vacuum drying, microwave drying, frying, vacuum freeze drying, hot air drying	–	–	54.25 mg/g	Vacuum drying	[152]
<i>Taraxacum mongolicum</i> leaf	Solar drying, hot air drying (50 °C, 65 °C, 80 °C, 95 °C), vacuum drying	10.77 mg/g	95 °C hot air drying	19.92 mg/g	Vacuum drying	[153]
Longan	Vacuum drying ∨ freeze drying ∨ hot air drying	2.855 mg/g	hot air drying	5.288 mg/g	Freeze drying	[154]
<i>Abelmoschus esculentus</i>	Hot air drying, vacuum freeze drying, microwave-hot air drying	–	–	3.03 µg/mL	Vacuum freeze drying	[155]
<i>Salvia miltiorrhiza</i>	Freeze drying, microwave drying, spray drying, vacuum drying	–	–	–	Freeze drying	[156]
<i>Lonicera japonica</i>	Hot air drying, shade drying, vacuum drying, vacuum freeze drying	12.43 mg/g	Hot air drying	37.74 mg/g	Vacuum freeze drying	[157]
Raspberry	Solar drying, hot air drying, heat pump drying, variable temperature differential pressure expansion drying, vacuum freeze drying	36.2 mg/g	Solar drying	84.2 mg/g	Vacuum freeze drying	[158]
<i>Jasminum sambac</i>	Hot air drying, microwave drying, hot air-microwave drying, vacuum freeze drying	15.02 mg/g	Hot air drying	16.21 mg/g	Vacuum freeze drying	[159]
<i>Burdock slices</i>	Hot air drying, vacuum microwave drying, vacuum freeze drying, hot air-vacuum microwave drying, vacuum freeze-vacuum microwave drying	–	Hot air drying	–	Vacuum freeze drying	[133]

the presence of oxygen, leading to their breakdown. Elevated temperatures further expedite the release of phenolic substances from their conjugated state, promoting their transformation into alternative compounds. Conversely, lower temperatures have been observed to mitigate the thermal degradation of phenolic substances. In another study, Tan et al.^[154] investigated the effects of vacuum drying, freeze drying and hot air drying on longan phenolic components. Their results indicated that freeze drying yielded the highest content of polyphenols. This outcome may be attributed to the susceptibility of longan phenolic substances to oxidized and decomposed at high temperatures, leading to a reduction in their content. Additionally, Zhang et al.^[133] studied the impact of five drying methods, including hot air drying, vacuum microwave drying, vacuum freeze drying, hot air combined with vacuum microwave drying, and vacuum freeze combined with vacuum microwave drying, on the phenolic substances in burdock slices. Their findings showed that vacuum freeze drying resulted in the highest content of polyphenols. This can be attributed to the conditions of vacuum and low temperature, which prevent heat-unstable polyphenols from coming into contact with oxygen, thereby preserving the phenolic substances effectively.

Indeed, despite variations in types, certain Chinese medicinal herbs exhibit the highest polyphenol content under hot air drying conditions. The increase in drying temperature triggers an enhancement in both antioxidant content and activity, thereby promoting the synthesis of numerous novel phenolic compounds during high-temperature drying. Consequently, this phenomenon results in an elevation in total phenolic content.

Ning et al.^[160] explored the efficacy of solar drying, hot air drying, and far infrared drying methods for red ginseng, discovering that the highest total phenolic content was achieved through hot air drying. This observation is attributed to the Maillard reaction occurring during high-temperature drying, which facilitated the conversion of bound polyphenols into free polyphenols, formation of novel phenolic compounds, or transformation of macromolecular polyphenolic compounds into lower molecular weight counterparts. Consequently, an increase in total phenolic content is observed. In another study, Qi et al.^[161] found that during hot air drying, the leaf portion of cress initially exhibited an higher polyphenol content, which underwent degradation at high temperatures throughout the drying process, resulting in a reduction in the final content. Conversely, the

stem portion predominantly synthesized polyphenols and demonstrated a significant increase in carvacrol and apigenin levels. Furthermore, Deng et al.^[162] investigated the total polyphenol content of *Phyllanthus emblica* treated with different drying methods and temperatures. They found that the content of *Phyllanthus emblica* obtained through vacuum freezing was lower. In the hot air drying group, the total polyphenol content significantly increased with the temperature, indicating that hot air drying at 80 °C was more conducive to the release of polyphenolic compounds in *Phyllanthus emblica* without reducing its antioxidant activity.

These studies suggest that the different drying methods exert varying impacts on the total phenolic content in various Chinese medicinal herbs, likely due to the intricate composition and differences in physicochemical properties among the herbs. However, the specific reasons underlying these observed effects remain to be further explored.

3.3. Flavonoid content

Flavonoids represent a class of naturally occurring bioactive compounds renowned for their diverse and intricate pharmacological properties, including antioxidant, anti-neoplastic, antibacterial, antiviral, and immunomodulatory activities, among others. Within this class, a broad range of flavonoids with complex structures exists, some of which are still undergoing investigation regarding their intricate structure-activity relationship and pharmacokinetic characteristics.^[163] Through an extensive review of relevant literature, it becomes apparent that the influence of drying methods on the flavonoid content in Chinese medicinal herbs demonstrates intricate and diverse characteristics, thereby hindering the formulation of definitive conclusions. Nevertheless, it is observed that common medicinal materials are well-suited for vacuum freeze drying and low-temperature drying. This preference is attributed to the susceptibility of flavonoids to structural alterations under high temperature conditions, where hydrolysis is prone to occur. Consequently, flavonoids may form new compounds upon interaction with water molecules, leading to the loss of original biological activity and a reduction in pharmacological effects. Furthermore, flavonoids are particularly susceptible to oxidation and degradation under elevated temperatures, thus significantly impacting their biological activity and stability. Hence, the choice of drying method plays a crucial role in preserving the

Table 7. Research on flavonoid contents in flowery herbs.

Herbs	Drying method	Flavonoid content(minimum)	Corresponding drying method	Flavonoid content(maximum)	Corresponding drying method	References
<i>Abelmoschus esculentus flower</i>	50 °C hot air drying, 70 °C hot air drying, microwave drying, vacuum freeze drying	–	70 °C hot air drying	–	Vacuum freeze drying	[164]
<i>Nelumbo nucifera</i>	Shade drying, solar drying, hot air drying, microwave drying, freeze drying	0.39 mg/g	Solar drying	60.20 mg/g	Freeze drying	[165]
<i>Paeonia suffruticosa</i>	Solar drying, hot air drying, microwave drying, vacuum freeze drying	7.98 mg/g	Microwave medium low temperature drying	11.85 mg/g	Vacuum freeze drying	[166]
<i>Albizia julibrissin Durazz.</i>	Solar drying, shade drying, steam bleaching drying, microwave killing drying, hot air drying (30 °C, 50 °C, 70 °C, 90 °C), freeze drying	–	Shade drying	–	Freeze drying	[167]
<i>Dendrobium officinale Kimura flower</i>	Solar drying, shade drying, hot air drying (60 °C, 80 °C, 100 °C), infrared drying, microwave drying, freeze drying	11.52 mg/g	Shade drying	15.66 mg/g	Freeze drying	[168]
<i>Lilium brownii</i>	Hot air drying(40 °C, 50 °C, 60 °C, 70 °C, 80 °C, 90 °C, 100 °C), solar drying, shade drying, vacuum drying, freeze drying	1.63 mg/g	40 °C hot air drying	3.84 mg/g	Freeze drying	[169]

Table 8. Research on flavonoid contents in leafy herbs.

Herbs	Drying method	Flavonoid yield/ content (minimum)	Corresponding drying method	Flavonoid yield/ content(maximum)	Corresponding drying method	References
<i>Angelica keiskei koidzumi</i>	Shade drying, hot air drying, vacuum freeze drying	–	Hot air drying	–	Vacuum freeze drying	[170]
<i>Morus alba L.</i>	Freeze drying, shade drying, hot air drying, solar drying, microwave drying, infrared drying	–	Infrared drying	–	Freeze drying	[171]
<i>Elaeagnus angustifolia L.</i>	Solar drying, shade drying, hot air drying, freeze drying	1.18%	Solar drying	1.62%	Freeze drying	[172]
<i>Epimedium brevicomu Maxim</i>	Freeze drying, shade drying, microwave, frying, shade drying after steaming, hot air drying	9.06%	200 °C hot air drying	10.39%	Freeze drying	[173]
<i>Scutellaria baicalensis</i>	Solar drying, shade drying, hot air drying, vacuum drying, microwave drying	193.31 mg/g	Shade drying	230.03 mg/g	Microwave drying	[174]
<i>Ginkgo biloba</i>	Solar drying, shade drying, hot air drying, microwave drying	–	Solar drying	–	Microwave drying	[175]

integrity and potency of flavonoids in Chinese medicinal herbs.

Most of the flowers of Chinese medicinal herbs are suitable for freeze drying, as shown in Table 7. Leaves of medicinal materials are best processed by vacuum freezing drying, followed by microwave drying, as shown in Table 8. For rhizome medicinal materials such as *Tetrastigma hemsleyanum*^[176] and *Radix puerariae*,^[177] freeze drying results in a high content of flavonoids. Conversely, *Panax notoginseng*^[178] and *Angelica sinensis*^[179] demonstrate an even higher

content of flavonoid when hot air dried. For fruit medicinal materials like *Chinese wolfberry*^[180] and *Crataegus pinnatifida*,^[181] freeze drying does not yield the highest flavonoid content. Instead, hot air drying or microwave drying is more suitable.

The above results also once again show that the types of Chinese medicinal herbs are complex, and it is necessary to continuously study the physical and chemical properties of Chinese medicinal herbs to find the best drying process and retain the medicinal ingredients to the maximum extent.

3.4. Saponin content

Saponins, a prominent class of phytochemicals found in Chinese medicinal herbs, constitute a distinctive group of glycosides that are widely distributed throughout nature. Clinical and pharmacological studies have demonstrated that saponins possess a range of beneficial effects, including anti-inflammatory, antibacterial, antiviral, and hypoglycemic.^[182] Moreover, their constituents exhibit potent anti-tumor properties, making saponins a focal point of research and development for novel anti-tumor drugs within the rapidly advancing field of Chinese medicinal herbs.^[183] Despite their minor side effects and ease of accessibility, it is crucial to conduct further investigation into the diverse range of saponins and their potential therapeutic applications. When evaluating saponins, it is essential to consider both their quantity and quality. However, it is important to exercise caution and avoid overlooking certain biologically inactive saponins, as their inclusion may affect the overall assessment of saponin content and efficacy. Therefore, a comprehensive approach to assessing saponins is warranted to ensure accurate evaluation and utilization of their therapeutic potential.

The use of mild drying conditions at low temperatures can effectively preserve the saponins and mitigate potential damage associated with high temperatures. Zhang et al.^[184] conducted a study employing nine different drying methods to dry *Paris polyphylla Smith var. yunnanensis* (Franch.) Hara and observed a decrease in the total saponin content with increasing heating power (temperature). This decline may be attributed to the destruction of endogenous enzymes under high-temperature conditions, leading to weakened respiration of the medicinal material itself and enhances diosgenin hydrolysis during extraction. Similarly, Wang et al.^[185] dried *Ophiopogon japonicas* (L. f.) Ker-Gawl at different temperatures and found that the saponin content was highest at 50 °C, significantly decreasing at 80 °C by more than 25%. This decline is speculated to result from high-temperature decomposition, while freeze drying at -70 °C also affected saponins. In another study, Li et al.^[186] dried ginseng flowers at varying temperatures and found that the optimal saponin quantity was obtained at 40 °C, with high-quality fractions decreasing as temperature increased. This suggests that high-temperature treatment may promote the transformation of ginseng flowers into rare saponins. Additionally, Xie et al.^[187] observed a significant reduction in the total saponin and saponin A content in the plant samples from the drying group compared

to the other groups, with the drying group exhibiting the lowest levels and the control group registering the highest levels. Notably, samples subjected to low-temperature drying at 55 °C exhibited the highest total saponin levels. This is attributed to the thermal inactivation of endogenous enzymes during high-temperature drying, preserving the efficacy of certain saponins while potentially compromising overall integrity and composition.

For the medicinal material of *polygala tenuifolia*, low-temperature drying does not yield optimal results due to its unique chemical and physical properties. Peng et al.^[188] discovered that the saponin content in *Polygala tenuifolia* root bark was the highest in microwave drying, followed by freeze drying, and the lowest in hot air drying at 40 °C. Additionally, the content of asiatic acid and saponins in *Polygala tenuifolia* root was the highest during microwave drying, followed by hot air drying at 60 °C; with no statistically significant difference observed between these two methods. Conversely, there were no significant differences found among the remaining six drying methods. Similarly, low-temperature drying is not suitable for *Panax notoginseng* and ginseng. Wu et al.^[178] dried *Panax notoginseng* using hot air drying at 50 °C and 80 °C, as well as freeze drying. They observed that the total saponin content in *Panax notoginseng* dried by 80 °C hot air drying was significantly lower than that in freeze drying. Moreover, compared to *Panax notoginseng* dried by 60 °C hot air drying, the contents of R_{g1} and R_{b1} in freeze-dried *Panax notoginseng* increased significantly by 8.37% and 6.33%, respectively, while the content of R₁ was decreased by 2.48%. Additionally, the contents of R_e and R_d in freeze-dried *Panax notoginseng* were also higher than those in hot air-dried *Panax notoginseng*. In another study, Lai et al.^[189] investigated the effects of three different drying processes – hot air drying, microwave drying and vacuum freeze drying – on the physical properties and saponins of ginseng. They found that the average sum of monomer saponins in vacuum freeze-dried ginseng was 7.6% and 26.03% higher than that in microwave-dried and hot air-dried ginseng, respectively. These results showed that vacuum freeze drying better preserves the morphology and color of ginseng, with the highest content of ginsenoside observed in vacuum freeze-dried ginseng.

The influence of drying temperature on saponin content is a complex process that requires comprehensive consideration of various factors, including temperature, humidity, and wind speed. Moreover, since saponins in Chinese medicinal herbs exhibit

differences in their composition and properties, future research should delve deeper into elucidating the impact of these factors on total saponin content to optimize the utilization of their medicinal properties. Understanding the interplay between drying conditions and saponin content is essential for maximizing the efficacy of Chinese medicinal herbs. By conducting systematic studies that examine how temperature, humidity, and other environmental factors affect saponin composition, researchers can develop more precise drying techniques that preserve saponin content and enhance the therapeutic potential of medicinal herbs.

Ultimately, by gaining a deeper understanding of the relationship between drying conditions and saponin content, researchers can contribute to the development of more effective strategies for processing Chinese medicinal herbs, thereby maximizing their medicinal benefits and promoting their widespread use in traditional and modern medicine.

3.5. Volatile oils content

Volatile oils are ubiquitous natural compounds possessing a diverse range of pharmacological activities, rendering them extensively employed in traditional Chinese medicine, aromatherapy, cosmetics, and other domains. It exhibits noteworthy antibacterial, anti-inflammatory, antioxidant, anti-tumor properties alongside transdermal absorption and analgesic effects. There are numerous Chinese medicinal herbs that contain volatile oils, particularly those from the *Lauraceae* family (such as mint and perilla), *Apiaceae* family (such as *angelicae sinensis* and fennel), *Compositae* family (such as *Artemisia argyi* and *Atractylodes lancea*), and *Autaceae* family (such as orange and pricklyash).^[190] These volatile oils play a significant role in traditional medicine practices and are increasingly recognized for their therapeutic

potential in modern healthcare applications. Their diverse pharmacological activities make them valuable ingredients in various formulations aimed at promoting health and well-being.

The effects of different drying methods on the content of volatile oils in Chinese medicinal herbs are summarized in Table 9. It is evident that for the production of volatile oils, priority should be given to low-temperature drying due to its shorter processing time and lower loss of volatile oils compared to solar drying and shade drying.

The diverse physiological properties of Chinese medicinal herbs necessitate careful consideration of drying conditions, as the selection of an appropriate drying method is critical for preserving the quality and efficacy of the final product. For certain medicinal materials rich in polysaccharides, high-temperature drying methods like microwave drying and hot air drying should be avoided to suppress enzyme degradation and retain polysaccharides. Instead, low-temperature drying is preferred. In contrast, for materials rich in polyphenols, low temperature or vacuum drying methods are preferred to inhibit the oxidation and decomposition of phenolic components. However, some materials benefit from high-temperature drying as it can increase antioxidant content and facilitate the synthesis of novel phenolic substances, thereby increasing total phenolic content. Most materials containing flavonoids are best suited for vacuum freeze drying to preserve optimal flavonoid content. Flavonoids are susceptible to hydrolysis, oxidation, and decomposition under high temperatures, resulting in reduced concentration; Saponins, being thermolabile compounds, are suitable for low-temperature drying to prevent decomposition and conversion into rare saponins. However, low-temperature drying may also lead to hydrolysis reactions and structural damage; The volatile oils are fully retained at low temperatures, high temperatures can lead to degradation and

Table 9. Research on volatile oils of Chinese medicinal herbs.

Herbs	Drying method	Volatile oils content (minimum)	Corresponding drying method	Volatile oils content (minimum)	Corresponding drying method	References
<i>Eupatorium fortunei</i>	50 °C hot air drying, solar drying, shade drying	29%	Solar drying	35%	50 °C hot air drying	[191]
<i>Cymbopogon citrates</i>	Solar drying, shade drying, 40 °C hot air drying	1.19%	Shade drying	1.27%	40 °C hot air drying	[192]
<i>Curcum a wenyujin</i>	Solar drying, shade drying, 30 °C hot air drying	0.74%	Solar drying	1.35%	30 °C hot air drying	[193]
<i>Sarcandra glabra leaf</i>	Solar drying, 50 °C hot air drying	0.10 mL·(100g) ⁻¹	Solar drying	0.20 mL·(100g) ⁻¹	50 °C hot air drying	[194]
<i>Angelica dahurica</i>	(45 °C, 50 °C, 60 °C, 70 °C) hot air drying	–	70 °C hot air drying	–	50 °C hot air drying	[195]
	(45 °C, 50 °C, 60 °C, 70 °C) medium wave infrared drying	–	70 °C medium wave infrared drying	–	45 °C medium wave infrared drying	

volatilization of volatile components. Although high temperatures are beneficial to the rapid evaporation of water and shorten the drying cycle, they also lead to the degradation and volatilization of volatile components.

The drying process will cause the physical effect of the change of the content of active components of Chinese medicinal materials. (1) Shrinkage of medicinal materials: As the material dries, water evaporates, leading to a reduction in volume and loss of elasticity in the tissue cells. This shrinkage can result in structural changes that affect the preservation and release of bioactive ingredients. (2) Surface hardening: High drying temperatures can cause the surface of the material to harden, impacting its permeability and extending drying time, potentially leading to greater loss of active ingredients. (3) Environmental factors such as light and oxygen: Drying materials in a light-free vacuum environment is beneficial for preserving active ingredients.

The findings presented above underscore the complexity of Chinese medicinal herbs, highlighting the need for ongoing research into their physical and chemical properties. This continuous investigation is crucial for identifying the optimal drying processes that maximize the retention of medicinal ingredients. Given the diverse array of Chinese medicinal herbs and their varied responses to different drying methods, a comprehensive understanding of their properties is essential. Through systematic analysis and experimentation, researchers can strive to develop drying techniques that preserve the medicinal efficacy of these herbs to the fullest extent possible.

Ultimately, by delving deeper into the physical and chemical characteristics of Chinese medicinal herbs, we can refine drying processes and ensure the preservation of their valuable medicinal components, thereby enhancing their therapeutic potential.

4. Discussion and future prospects

The drying treatment is a crucial step in the preservation process of Chinese medicinal herbs, as it not only impacts the sensory attributes such as color, odor, and taste, but also directly influences their medicinal value and clinical efficacy. The process of drying effectively eliminates a significant amount of moisture from the medicinal materials, thereby preventing the proliferation of molds, bacteria, and other microorganisms. This ensures the optimal quality of the Chinese medicinal herbs. In the context of global energy scarcity and the imperative for energy conservation and emission reduction, it is significant for China to develop

drying equipment and technology that align with the carbon dioxide mitigation objective, thereby achieving enhanced sustainability. Therefore, the development of energy-efficient drying equipment becomes imperative. This paper provides a comprehensive overview of the drying technology, energy consumption, and medicinal components of Chinese medicinal herbs. Furthermore, it presents an analysis of the challenges encountered during the drying process of Chinese medicinal herbs as follows:

1. Currently, a range of drying technologies have been employed for the dehydration of Chinese medicinal herbs. However, certain drying techniques cannot be integrated into the large-scale drying industry of Chinese medicinal herbs due to challenges in equipment operation control and high capital investment costs.
2. Combination drying is achieved by integrating two or more individual drying methods to accentuate their respective merits. The dried medicinal products obtained through combined drying exhibit enhancements in terms of active ingredients, equipment energy consumption, drying rate, color, taste, water content and other aspects.
3. In terms of composition, the majority of components found in Chinese medicinal herbs are heat-sensitive. The utilization value of these materials can be significantly reduced due to the substantial loss of their content caused by high temperature drying. Therefore, it is crucial to select appropriate drying methods and establish optimal process parameters in order to ensure the quality and efficacy of medicinal materials.
4. Vacuum freeze drying is the most effective low temperature drying technologies, however, it entails high investment costs, energy consumption, and lacks industrial scalability. On the other hand, high voltage electric field drying exhibits promising prospects as a non-thermal drying technology, nevertheless, further research is required to elucidate its drying mechanism and optimize process parameters.
5. The determination of optimal drying process parameters necessitates a comprehensive investigation into the intricate physical and chemical properties of Chinese medicinal herbs, as well as an exploration of the fundamental principles underlying the drying process.

Looking ahead, the future development trend of drying technology for Chinese medicinal herbs

primarily involves integrating various individual drying processes to leverage their respective advantages in order to enhance drying efficiency and reduce energy consumption; develop versatile equipment for meeting the drying requirements of diverse Chinese medicinal herbs, enhancing the adjustable range of temperature, air humidity, wind speed, etc.; develop large-scale and specialized drying equipment for application in the extensive Chinese medicinal herbs drying industry; achieve fully automated control or remote monitoring of the drying processes. Therefore, it is imperative to conduct further research and optimize the drying techniques employed for Chinese medicinal herbs. This endeavor not only enhances the quality of Chinese medicinal herbs, but also fosters the flourishing development of traditional Chinese medicine.

Authors' contributions

X.L.: writing – original draft preparation, data curation, investigation; Y.W.: conceptualization, funding acquisition, writing – review & editing, supervision; X.M.: funding acquisition; M.L.: investigation, supervision; D.K.: conceptualization; P.P.: formal analysis; Q.W.: visualization; T.K.A.: formal analysis. All authors have read and agreed to the published version of the manuscript.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was supported by the National Natural Science Foundation of China (Grant No. 52376211), the Major Science and Technology Special Project of Yunnan Province (Grant No. 202202AE090011), Young and Middle-aged Academics and Technical Leaders Training Project of Yunnan Province (Grant No.202305AC160085).

Data availability statement

No new data were created or analyzed in this study. Data sharing is not applicable to this article.

References

- [1] Hu, Y.; Liu, P.; Xiong, H.; Wang, W.; Zhang, S.; Ma, X.; He, Y.; Yan, K.; Yan, X. Digital & Intelligent Chinese Materia Medica: Upgrade and Innovation of Modern Chinese Materia Medica. *Chin. Trad. Herbal Drugs*. **2024**, *55*, 1–11. DOI: [10.7501/j.issn.0253-2670.2024.01.001](https://doi.org/10.7501/j.issn.0253-2670.2024.01.001).
- [2] Wu, L.; Miao, Z.; Wei, N.; Man, D.; Li, Y.; Xu, K. Development Status and Exploration of Chinese Herbal Medicine Industry Mechanization Development in China. *Forest. Mach. Woodwork. Equip.* **2021**, *49*, 13–18. DOI: [10.13279/j.cnki.fmwe.2021.0110](https://doi.org/10.13279/j.cnki.fmwe.2021.0110).
- [3] Wang, C.; Li, Z. Current Situation and Future Trend of Traditional Chinese Medicine Innovation Development. *Chin. J. New Drugs*. **2023**, *32*, 2013–2021.
- [4] Jin, J.; Li, T. Research on Comprehensive Evaluation of the Development Level of Chinese Traditional Medicine Culture Soft Power. *Enterp. Hebei*. **2024**, *1*, 5–9. DOI: [10.19885/j.cnki.hbqy.2024.01.018](https://doi.org/10.19885/j.cnki.hbqy.2024.01.018).
- [5] Cai, W. Application Status and Development Prospect of Pediatric Traditional Chinese Medicine Technology. *J. Trad. Chin. Med. Manage.* **2023**, *31*, 238–240. DOI: [10.16690/j.cnki.1007-9203.2023.13.018](https://doi.org/10.16690/j.cnki.1007-9203.2023.13.018).
- [6] Yang, C. Progress and Prospect of Application of Traditional Chinese Medicine in Prevention and Treatment of Influenza. *J. Trad. Chin. Med. Manage.* **2023**, *31*, 226–228. DOI: [10.16690/j.cnki.1007-9203.2023.06.078](https://doi.org/10.16690/j.cnki.1007-9203.2023.06.078).
- [7] Li, J.; Zhou, Y.; Li, X.; Mao, S.; Shen, S.; Tian, S.; Zhang, P.; Song, S.; Fu, W.; Zhu, H. Application Value and Development Prospect of TCM Clinical Big Data Knowledge Engineering. *Lishizhen Med. Mater. Medica Res.* **2023**, *34*, 769–770. DOI: [10.3969/j.issn.1008-0805.2023.03.75](https://doi.org/10.3969/j.issn.1008-0805.2023.03.75).
- [8] Huang, W. Study on Preservation Technology of Chinese Medicine Decoction Pieces. *World Latest Med. Inf.* **2016**, *16*, 206. DOI: [10.3969/j.issn.1671-3141.2016.3.153](https://doi.org/10.3969/j.issn.1671-3141.2016.3.153).
- [9] Li, J. Application of Drying Technology in Primary Processing of Chinese Medicinal Materials. *Agric. Eng. Technol.* **2020**, *40*, 91. DOI: [10.16815/j.cnki.11-5436/s.2020.23.065](https://doi.org/10.16815/j.cnki.11-5436/s.2020.23.065).
- [10] Li; Qiu, X.; Qi, Y.; Wan, Q.; Gao; Wang, X.; Yang, M.; Li, Y. Research Status of the Change Mechanism of Active Components in Chinese Medicinal Materials during Drying. *China J. Chin. Mater. Med.* **1970**, 1–13. DOI: [10.19540/j.cnki.cjcmm.20231115.301](https://doi.org/10.19540/j.cnki.cjcmm.20231115.301).
- [11] Fan, T.; Dou, Z.; Li, J.; Wen, Y.; Liu, L.; Liu, Y. Effects of Different Drying Methods on Chemical Components of Traditional Chinese Medicine: A Review. *Modern Chin. Med.* **2021**, *23*, 2017–2024. DOI: [10.13313/j.issn.1673-4890.20200824004](https://doi.org/10.13313/j.issn.1673-4890.20200824004).
- [12] Yu, F.; Wan, N.; Li, Y.; Wang, X.; Wu, Z.; Liu, Z.; Yang, M. Analysis on Change Rule and Mechanism in Physical and Chemical Properties of Chinese Herbal Medicines during Drying. *Chin. Tradit. Herbal Drugs*. **2021**, *52*, 2144–2153. DOI: [10.7501/j.issn.0253-2670.2021.07.031](https://doi.org/10.7501/j.issn.0253-2670.2021.07.031).
- [13] Zhu, W.; Zhao, Y.; Wu, J.; Chen, P.; Jiang; Chen, N.; Bao, H.; Wang, W. Research Progress on Drying Technology and Equipment of Rhizome Traditional Chinese Medicines. *J. China Agric. Univ.* **2023**, *28*, 153–171. DOI: [10.11841/j.issn.1007-4333.2023.01.13](https://doi.org/10.11841/j.issn.1007-4333.2023.01.13).
- [14] Zhang, X. Present Situation and Development Trend of Drying Technology of Chinese Medicinal

- Materials. *Elect. J. Clin. Med. Lit.* **2020**, *7*, 194. DOI: [10.16281/j.cnki.jocml.2020.34.156](https://doi.org/10.16281/j.cnki.jocml.2020.34.156).
- [15] Wang, L.; Li, C.; Liu, M.; Li, K.; Ye, X.; Wang, Z.; Fan, L.; Wang, R.; Zhao, H.; Kan, Z.; et al. Research Status of Drying Technology and Equipment of Chinese Medicinal Materials. *Transact. Chin. Soc. Agric. Eng.* **1970**, 1–27. DOI: [10.11975/j.issn.1002-6819.202306104](https://doi.org/10.11975/j.issn.1002-6819.202306104).
- [16] Zhao, C. Research and Development Trend of Drying Equipment Technology for Chinese Medicinal Materials. *J. Mudanjiang Normal Univ. (Nat. Sci. Ed.)*. **2015**, 18–20. DOI: [10.13815/j.cnki.jmtc\(ns\).2015.02.009](https://doi.org/10.13815/j.cnki.jmtc(ns).2015.02.009).
- [17] Xie, H.; Qi, Y.; Wan, N.; Wu, Z.; Wang, X.; Li, Y.; Yang, M. Shrinkage Mechanism, Influencing Factors and Control Strategies in Drying Process of Chinese Medicinal Materials. *Chin. Tradit. Herbal Drugs*. **2022**, *53*, 2872–2881. DOI: [10.7501/j.issn.0253-2670.2022.09.031](https://doi.org/10.7501/j.issn.0253-2670.2022.09.031).
- [18] Polatci, H.; Taşova, M.; Şin, B. Effects of Pre-Treatments on Drying Kinetics and Energy Consumption, Heat-Mass Transfer Coefficients, Micro-Structure of Jujube (*Zizyphus jujuba* L.) Fruit. *Food Sci. Technol.* **2022**, *42*, e15721. DOI: [10.1590/fst.15721](https://doi.org/10.1590/fst.15721).
- [19] Li, Y.; Zhu, W.; Chen, P.; Wang, Y.; Jiang, Wang, D.; Chen, L.; Zhang, Y.; Hu, X.; Wang, J. Study on Hot Air Drying Characteristics and Thermodynamic Properties of Different Varieties of High Oleic Peanuts. *Sci. Tech. Food Indus.* **1970**, 1–16. DOI: [10.13386/j.issn1002-0306.2023070193](https://doi.org/10.13386/j.issn1002-0306.2023070193).
- [20] Hunashikatti, P. T.; Ramaswamyreddy, S.; Sethia, P. M.; Goyal, S.; Rohan, S. K.; Raj, S. Experimental Studies on Thin-Layer Drying of Mint Leaves in a Solar Dryer and under Open Sun. *Current Sci.* **2022**, *122*, 1066. DOI: [10.18520/cs/v122/i9/1066-1071](https://doi.org/10.18520/cs/v122/i9/1066-1071).
- [21] Husham Abdulmalek, S.; Khalaji Assadi, M.; Al-Kayiem, H. H.; Gitan, A. A. A Comparative Analysis on the Uniformity Enhancement Methods of Solar Thermal Drying. *Energy*. **2018**, *148*, 1103–1115. DOI: [10.1016/j.energy.2018.01.060](https://doi.org/10.1016/j.energy.2018.01.060).
- [22] Sharshir, S. W.; Joseph, A.; Peng, G.; Kandeal, A. W.; Abdullah, A. S.; Abdelaziz, G. B.; Edreis, E. M. A.; Yuan, Z. Recent Efforts in Developing Agricultural Product Drying Processes Using Solar Energy. *Sol. Energy*. **2023**, *257*, 137–154. DOI: [10.1016/j.solener.2023.04.022](https://doi.org/10.1016/j.solener.2023.04.022).
- [23] Chen, L.; Sun, J.; Zhu, X.; Wang, F.; Fan, B.; Wang, Y.; Huang, Y.; Zhang, F.; Li, M.; Hu, S.; et al. Changes of Chemical Constituents of *Angelica Sinensis* Diels during Different Processing. *Food Res. Develop.* **2023**, *44*, 209–216. DOI: [10.12161/j.issn.1005-6521.2023.023.029](https://doi.org/10.12161/j.issn.1005-6521.2023.023.029).
- [24] Deng, H.; Gao, B.; Wang, X. Li. Effect of Drying Method on the Content of Active Components of Chinese Herbal Medicine *Salvia Miltiorrhiza* Bunge. *J. Shangluo Univ.* **2007**, *21*, 54–56. DOI: [10.13440/j.slxy.1674-0033.2007.02.017](https://doi.org/10.13440/j.slxy.1674-0033.2007.02.017).
- [25] Slama, R. B.; Combarous, M. Study of Orange Peels Drying Kinetics and Development of a Solar Dryer by Forced Convection. *Sol. Energy*. **2011**, *85*, 570–578. DOI: [10.1016/j.solener.2011.01.001](https://doi.org/10.1016/j.solener.2011.01.001).
- [26] Abdelkader, T. K.; Salem, A. E.; Zhang, Y.; Gaballah, E. S.; Makram, S. O.; Fan, Q. Energy and Exergy Analysis of Carbon Nanotubes-Based Solar Dryer. *J. Energy Storage*. **2021**, *39*, 102623. DOI: [10.1016/j.est.2021.102623](https://doi.org/10.1016/j.est.2021.102623).
- [27] Abdelkader, T. K.; El. Salem, A.; Zhang, Y.; Gaballah, E. S.; Refai, M.; Torki, M.; Fan, Q. Flat Micro Heat Pipe-Based Shell and Tube Storage Unit for Indirect Solar Dryer: A Pilot Study. *Environ. Sci. Pollut. Res.* **2023**, *31*, 46385–46396. DOI: [10.1007/s11356-023-27851-z](https://doi.org/10.1007/s11356-023-27851-z).
- [28] Kamarulzaman, A.; Hasanuzzaman, M.; Rahim, N. A. Global Advancement of Solar Drying Technologies and Its Future Prospects: A Review. *Sol. Energy*. **2021**, *221*, 559–582. DOI: [10.1016/j.solener.2021.04.056](https://doi.org/10.1016/j.solener.2021.04.056).
- [29] Lakshmi, D. V. N.; Muthukumar, P.; Layek, A.; Nayak, P. K. Drying Kinetics and Quality Analysis of Black Turmeric (*Curcuma Caesia*) Drying in a Mixed Mode Forced Convection Solar Dryer Integrated with Thermal Energy Storage. *Renew. Energy*. **2018**, *120*, 23–34. DOI: [10.1016/j.renene.2017.12.053](https://doi.org/10.1016/j.renene.2017.12.053).
- [30] Mehta, P.; Bhatt, N.; Bassan, G.; Kabeel, A. E. Performance Improvement and Advancement Studies of Mixed-Mode Solar Thermal Dryers: A Review. *Environ. Sci. Pollut. Res. Int.* **2022**, *29*, 62822–62838. DOI: [10.1007/s11356-022-21736-3](https://doi.org/10.1007/s11356-022-21736-3).
- [31] China Wind and Solar Energy Resources Annual Outlook Bulletin. *Chin. Energy News*. **2022**, 013.
- [32] Salehi, F. Recent Applications of Heat Pump Dryer for Drying of Fruit Crops: A Review. *Int. J. Fruit Sci.* **2021**, *21*, 546–555. DOI: [10.1080/15538362.2021.1911746](https://doi.org/10.1080/15538362.2021.1911746).
- [33] Mao, Y.; Li, Y. Research Progress of Heat Pump Drying Technology. *Chem. Eng. Equipment*. **2023**, *12*, 184–185 + 193. DOI: [10.19566/j.cnki.cn35-1285/tq.2023.12.054](https://doi.org/10.19566/j.cnki.cn35-1285/tq.2023.12.054).
- [34] Zhang, P.; Wu, X.; Zhang, Z.; Yue, Y.; Yang, J. Heat Pump Drying Technology and Its Application Prospects in Agricultural Special Products. *Refriger Air-Condition*. **2019**, *19*, 65–71.
- [35] Ren, X.; Chen, Q.; Gao, Y.; Wang, L.; Guo, J.; Guo, X. Research Progress of Heat Pump Drying Technology. *Energy Conserv.* **2021**, *40*, 74–76. DOI: [10.3969/j.issn.1004-7948.2021.04.023](https://doi.org/10.3969/j.issn.1004-7948.2021.04.023).
- [36] Zhang, Y.; Chen, P.; Chen, J.; Li, G.; Lin, Y.; Li, X.; Yang, N.; Hao, H. Application Status and Development Trend of Heat Pump Drying Technology for Agricultural Products under the Dual Carbon Target. *J. Anhui Agric. Sci.* **2023**, *51*, 12–16 + 35. DOI: [10.3969/j.issn.0517-6611.2023.23.004](https://doi.org/10.3969/j.issn.0517-6611.2023.23.004).
- [37] Zhang, L.; Zhang, X.; Fan, Y.; Chen, S.; Zhang, C. Research Progress of Heat Pump Drying Technology and Its Application in Drying of Shiitake Mushrooms. *Refriger Air-Condition*. **2019**, *19*, 77–83 + 87.
- [38] Lv, D.; Jiang, Z. Influence of Circulating Air Volume of the Refrigeration System on

- Dehumidification Performance of Heat Pump Drying. *Chin. Forest Prod. Indus.* **2022**, 59, 40–45. DOI: [10.19531/j.issn1001-5299.202208007](https://doi.org/10.19531/j.issn1001-5299.202208007).
- [39] Wang, Z.; Zhang, L.; Xu, X.; Wu, Q. Experimental Research on Heat Pump Drying System for Medicinal Herbs Storage. *Pack. Eng.* **2024**, 45, 1–9. DOI: [10.19554/j.cnki.1001-3563.2024.01.001](https://doi.org/10.19554/j.cnki.1001-3563.2024.01.001).
- [40] Xu, C.; Tong, T.; Han, C.; Jiang, Y. Research on the Application of Microwave Drying Technology and Equipment for Crop Seeds. *Agricult. Mach. Mainten.* **2023**, 41–43. DOI: [10.14031/j.cnki.jnwjx.2023.06.012](https://doi.org/10.14031/j.cnki.jnwjx.2023.06.012).
- [41] Wang, Q.; Gu, Z.; Bai, J.; Liu, J.; Liu, H. Comparative Study of the Characteristics of Oil Shale with Hot Air Drying and Microwave Drying. *Energy Procedia.* **2012**, 17, 884–891. DOI: [10.1016/j.egypro.2012.02.183](https://doi.org/10.1016/j.egypro.2012.02.183).
- [42] Zheng, Y.; Xie, M.; Zhang, F.; Li, Y.; Li, M. Drying Technology Applied in Processing of Chinese Medicinal Materials. *Gansu Agricult. Sci. Technol.* **2017**, 3, 71–74. DOI: [10.3969/j.issn.1001-1463.2017.03.022](https://doi.org/10.3969/j.issn.1001-1463.2017.03.022).
- [43] Zhao, H.; Wang, Y.; Ma, Y.; Liu, Y.; Wen, J.; Zhang, B. Effects of Different Microwave Drying Conditions on Drying Characteristics and Quality of Preserved Apricots. *Food Mach.* **2023**, 39, 123–129. DOI: [10.13652/j.spjx.1003.5788.2022.80874](https://doi.org/10.13652/j.spjx.1003.5788.2022.80874).
- [44] Wang, S.; Nie, Q.; Zhang, B.; Zhang, L.; Zang, C.; Wang, G.; Shang, Y. Application of Microwave Drying and Sterilization Technology in TCM Field. *Chin. J. Inform. TCM.* **2017**, 24, 132–136. DOI: [10.3969/j.issn.1005-5304.2017.11.034](https://doi.org/10.3969/j.issn.1005-5304.2017.11.034).
- [45] Lian, Y.; Qiu, M.; Qiu, H.; Jiang, G. Advances in Microwave Drying Technology for Chinese Medicinal Materials. *Asia-Pacific Tradit. Med.* **2017**, 13, 45–48. DOI: [10.11954/ytctty.201718016](https://doi.org/10.11954/ytctty.201718016).
- [46] Yang, Z. Study on Heat and Mass Transfer Law of Microwave Drying of Rice and Optimization of Control Method of the Device. Master Thesis, Anhui University of Technology, **2023**.
- [47] Beşir, A.; Gökmen, S.; Gül, L. B.; Yazıcı, F.; Gül, O. Evaluating of Microwave Drying for Hawthorn Slice as Alternative to Convective Drying. *Braz. Arch. Biol. Technol.* **2022**, 65, e22210614. DOI: [10.1590/1678-4324-2022210614](https://doi.org/10.1590/1678-4324-2022210614).
- [48] Wu, Z.; Li, X.; Meng, L.; Wei, Y.; Li, B.; Huang, J.; Huang, Z. Drying Characteristics of Chestnut Slices under Three Microwave Drying Modes. *Food Res. Develop.* **2023**, 44, 53–60 + 177. DOI: [10.12161/j.issn.1005-6521.2023.17.009](https://doi.org/10.12161/j.issn.1005-6521.2023.17.009).
- [49] Wu, Z.; Zhang, X.; Huang, J.; Li, H.; Sun, W.; Li, B.; Huang, Z. Characteristics and Model Fitting of Intermittent Microwave Drying of Polygonatum Sibiricum. *Jiangsu Agricult. Sci.* **2022**, 50, 156–162. DOI: [10.15889/j.issn.1002-1302.2022.15.025](https://doi.org/10.15889/j.issn.1002-1302.2022.15.025).
- [50] Jin, X. Study on Optimization of Microwave Drying Device. *Food Proces. Mach.* **2023**, 29, 8–10. DOI: [10.16736/j.cnki.cn41-1434/ts.2023.01.003](https://doi.org/10.16736/j.cnki.cn41-1434/ts.2023.01.003).
- [51] Du, Y.; Xie, H.; Wei, H.; Yan, J.; Xu, X.; Chen, W. Development Status of Far-Infrared Grain Drying Equipment. *J. Anhui Agricult. Mech.* **2022**, 43, 107–117. DOI: [10.13733/j.jcam.issn.2095-5553.2022.11.016](https://doi.org/10.13733/j.jcam.issn.2095-5553.2022.11.016).
- [52] Zhao, H.; Dai, J.; Qiao, L.; Zhang, L.; Wu, K. Far Infrared Drying of Sea Cucumber. *Modern Food Sci. Technol.* **2018**, 34, 101–107. DOI: [10.13982/j.mfst.1673-9078.2018.03.015](https://doi.org/10.13982/j.mfst.1673-9078.2018.03.015).
- [53] Jiang, C.; Wan, F.; Xie, Y.; Zhao, Y.; Li, W.; Qiu, J.; Huang, X. Optimization of Far-Infrared Drying Process of Stevia Rebaudiana Leaves Based on AHP-Entropy Weight Method. *J. Northwest A&F Univ. (Nat. Sci. Ed.)*. **2022**, 50, 101–109. DOI: [10.13207/j.cnki.jnwafu.2022.06.012](https://doi.org/10.13207/j.cnki.jnwafu.2022.06.012).
- [54] Pei, Y.; Li, Z.; Xu, W.; Song, C.; Li, J.; Song, F. Effects of Ultrasound Pretreatment Followed by Far-Infrared Drying on Physicochemical Properties, Antioxidant Activity and Aroma Compounds of Saffron (*Crocus Sativus* L.). *Food Biosci.* **2021**, 42, 101186. DOI: [10.1016/j.fbio.2021.101186](https://doi.org/10.1016/j.fbio.2021.101186).
- [55] Huang, X.; Li, W.; Wang, Y.; Wan, F. Drying Characteristics and Quality of Stevia Rebaudiana Leaves by Far-Infrared Radiation. *LWT.* **2021**, 140, 110638. DOI: [10.1016/j.lwt.2020.110638](https://doi.org/10.1016/j.lwt.2020.110638).
- [56] Shi, X.; Liu, Y.; Li, Z.; Sun, X.; Li, X. Effects of Radiation Temperature on Dehydration and Moisture Migration in Banana Slices during Far-Infrared Radiation Drying. *J. Food Process. Preserv.* **2020**, 44, 1–11. DOI: [10.1111/jfpp.14901](https://doi.org/10.1111/jfpp.14901).
- [57] Li, W.; Wan, F.; Luo, Y.; Wei, B.; Huang, X. Far Infrared Drying Characteristics and Kinetics of Angelicae Sinensis Radix Slices. *Chin. Trad. Herbal Drugs.* **2019**, 50, 4320–4328. DOI: [10.7501/j.issn.0253-2670.2019.18.010](https://doi.org/10.7501/j.issn.0253-2670.2019.18.010).
- [58] Zhou, S.; Duan, X.; Ren, G.; Zhang, M.; Ma, L.; Che, X. Effect of Thickness on Moisture Transfer during Far-Infrared Drying of Dioscorea oppositifolia. *Food Machin.* **2019**, 35, 75–81. DOI: [10.13652/j.issn.1003-5788.2019.12.014](https://doi.org/10.13652/j.issn.1003-5788.2019.12.014).
- [59] Zhang, Q.; Wan, F.; Zang, Z.; Jiang, C.; Xu, Y.; Huang, X. Effect of Ultrasonic Far-Infrared Synergistic Drying on the Characteristics and Qualities of Wolfberry (*Lycium Barbarum* L.). *Ultrason. Sonochem.* **2022**, 89, 106134. DOI: [10.1016/j.ultsonch.2022.106134](https://doi.org/10.1016/j.ultsonch.2022.106134).
- [60] Zhang, D.; Huang, D.; Zhang, Y.; Lu, Y.; Huang, S.; Gong, G.; Li, L. Ultrasonic Assisted Far Infrared Drying Characteristics and Energy Consumption of Ginger Slices. *Ultrason. Sonochem.* **2023**, 92, 106287. DOI: [10.1016/j.ultsonch.2022.106287](https://doi.org/10.1016/j.ultsonch.2022.106287).
- [61] Shang, J.; Zhang, Q.; Wang, T.; Xu, Y.; Zang, Z.; Wan, F.; Yue, Y.; Huang, X. Effect of Ultrasonic Pretreatment on the Far-Infrared Drying Process and Quality Characteristics of Licorice. *Foods.* **2023**, 12, 2414. DOI: [10.3390/foods12122414](https://doi.org/10.3390/foods12122414).
- [62] Rajoriya, D.; Shewale, S. R.; Bhavya, M. L.; Hebbar, H. U. Far Infrared Assisted Refractance Window Drying of Apple Slices: Comparative Study on Flavour, Nutrient Retention and Drying Characteristics. *Innovat. Food Sci. Emerg. Technol.* **2020**, 66, 102530. DOI: [10.1016/j.ifset.2020.102530](https://doi.org/10.1016/j.ifset.2020.102530).
- [63] Xie, L.; Mujumdar, A. S.; Fang, X.-M.; Wang, J.; Dai, J.-W.; Du, Z.-L.; Xiao, H.-W.; Liu, Y.; Gao, Z.-J. Far-Infrared Radiation Heating Assisted Pulsed

- Vacuum Drying (FIR-PVD) of Wolfberry (*Lycium Barbarum* L.): Effects on Drying Kinetics and Quality Attributes. *Food Bioprod. Process.* **2017**, *102*, 320–331. DOI: [10.1016/j.fbp.2017.01.012](https://doi.org/10.1016/j.fbp.2017.01.012).
- [64] Oliveira, N. L.; Silva, S. H.; Figueiredo, J. D. A.; Norcino, L. B.; Resende, J. V. D. Infrared-Assisted Freeze-Drying (IRFD) of Açai Puree: Effects on the Drying Kinetics, Microstructure and Bioactive Compounds. *Innov. Food Sci. Emerg. Technol.* **2021**, *74*, 102843. DOI: [10.1016/j.ifset.2021.102843](https://doi.org/10.1016/j.ifset.2021.102843).
- [65] Li, M.; Cao, Y. Design of Vacuum Drying System. *Technol. Equip.* **2023**, *49*, 89–91 + 110.
- [66] Wang, Z.; Zhang, W.; Cui, J. Research Progress and Prospect of Vacuum Drying Technology. *Chem. Eng. Mach.* **2021**, *48*, 321–325.
- [67] Xu, P.; Peng, X.; Yang, J.; Li, X.; Zhang, H.; Jia, X.; Liu, Y.; Wang, Z.; Zhang, Z. Effect of Vacuum Drying and Pulsed Vacuum Drying on Drying Kinetics and Quality of Bitter Orange (*Citrus Aurantium* L.) Slices. *J. Food Process. Preserv.* **2021**, *45*, 1–16. DOI: [10.1111/jfpp.16098](https://doi.org/10.1111/jfpp.16098).
- [68] Yan, Z.; Meng, Q. Study on Vacuum Drying Characteristics of *Cornus officinalis*. *Agric. Henan.* **2022**, 53–54. DOI: [10.15904/j.cnki.hnny.2022.09.026](https://doi.org/10.15904/j.cnki.hnny.2022.09.026).
- [69] Liu, X.; Zhao, L.; Wang, Y.; Zhang, J. Vacuum Drying Characteristics and Dynamic Simulation of Hawthorn. *Storage Process.* **2022**, *22*, 50–55.
- [70] Zhang, X.; Zhang, J.; Peng, G.; Zhang, H.; Luo, S.; Yang, L. Optimization of Vacuum Drying Technology of Yam Based on Multi-Objective Genetic Algorithm. *Chin. J. Vacuum Sci. Technol.* **2021**, *41*, 1101–1109. DOI: [10.13922/j.cnki.cjvst.202106010](https://doi.org/10.13922/j.cnki.cjvst.202106010).
- [71] Qian, J.; Zhang, Q.; Wang, J.; Fang, X.; Zhang, W.; Gao, Z.; Liu, Y.; Xiao, H. Effects of Three Drying Technologies on Drying Characteristics and Quality Attributes of Jujube Crisps. *Transact. Chin. Soc. Agric. Eng.* **2016**, *32*, 259–265.
- [72] Zhang, W.; Chen, H.; Fan, X.; Xiao, H.; Zheng, Z.; Ju, H. Pulsed Vacuum Drying Characteristics and Multi-Objective Optimization of Poria Cubes. *Food Mach.* **2021**, *37*, 91–98, 144. DOI: [10.13652/j.issn.1003-5788.2021.09.015](https://doi.org/10.13652/j.issn.1003-5788.2021.09.015).
- [73] Ren, H. Vacuum Freeze-Drying Technology and Its Application in the Field of Chinese Medicine. *Mech. Elect. Inform.* **2016**, 12–21. DOI: [10.19514/j.cnki.cn32-1628/tm.2016.20.003](https://doi.org/10.19514/j.cnki.cn32-1628/tm.2016.20.003).
- [74] Guo, J.; Zhang, G. Study on the Forsythia Suspensa Processing of Vacuum Freeze-Drying Technology. *Farm Products Proces.* **2022**, 16–20 + 25. DOI: [10.16693/j.cnki.1671-9646\(X\).2022.01.034](https://doi.org/10.16693/j.cnki.1671-9646(X).2022.01.034).
- [75] Tian, J.; Yao, Z.; Zhang, Z.; Zhang, Z.; Ji, H.; Wu, J. Effects of Sublimation Temperature on Cell Microstructure of Pear Melon during Vacuum Freeze Drying. *Food Mach.* **2023**, *39*, 12–17 + 44. DOI: [10.13652/j.spjx.1003.5788.2023.80313](https://doi.org/10.13652/j.spjx.1003.5788.2023.80313).
- [76] Yang, H.; Zhao, H.; Chen, Z.; Yin, Z.; Li, W. Study on the Process of Vacuum Freeze-Drying of Tianma (*Gastrodia Elata*). *Guid. J. Trad. Chin. Med. Pharm.* **2019**, *25*, 42–44. DOI: [10.13862/j.cnki.cn43-1446/r.2019.17.012](https://doi.org/10.13862/j.cnki.cn43-1446/r.2019.17.012).
- [77] Bi, J.; Feng, S.; Jin, X.; Yi, J.; Li, X.; Lv, J.; Wu, X. Development and Trend of Vacuum Freeze-Drying Technology and Industry. *J. Nuclear Agric. Sci.* **2022**, *36*, 414–421. DOI: [10.11869/j.issn.100-8551.2002.02.0414](https://doi.org/10.11869/j.issn.100-8551.2002.02.0414).
- [78] Mao, Y. Protocol Development of Radio Frequency Drying and Disinfestation for Walnuts. Master Thesis, Northwest Agriculture and Forestry University, **2023**.
- [79] Altemimi, A.; Aziz, S. N.; Al-Hilphy, A. R. S.; Lakhssassi, N.; Watson, D. G.; Ibrahim, S. A. Critical Review of Radio-Frequency (RF) Heating Applications in Food Processing. *Food Qual. Saf.* **2019**, *3*, 81–91. DOI: [10.1093/fqsafe/fyz002](https://doi.org/10.1093/fqsafe/fyz002).
- [80] Lian, F.; Sun, D.-W.; Cheng, J.-H.; Ma, J. Improving Modification of Structures and Functionalities of Food Macromolecules by Novel Thermal Technologies. *Trends Food Sci. Technol.* **2022**, *129*, 327–338. DOI: [10.1016/j.tifs.2022.10.001](https://doi.org/10.1016/j.tifs.2022.10.001).
- [81] Guan, X.; Lin, B.; Xu, Y.; Yang, G.; Xu, J.; Zhang, S.; Li, R.; Wang, S. Recent Developments in Pasteurising Seeds and Their Products Using Radio Frequency Heating: A Review. *Int. J. Food Sci. Tech.* **2022**, *57*, 3223–3243. DOI: [10.1111/ijfs.15497](https://doi.org/10.1111/ijfs.15497).
- [82] Jiao, Y.; Tang, J.; Wang, Y.; Koral, T. L. Radio-Frequency Applications for Food Processing and Safety. *Annu. Rev. Food Sci. Technol.* **2018**, *9*, 105–127. DOI: [10.1146/annurev-food-041715-033038](https://doi.org/10.1146/annurev-food-041715-033038).
- [83] Li, L. Design and Testing of Figs Vacuum Pulsating Radio Frequency Blanching and Drying Device. Master Thesis, Sichuan Agricultural University, **2023**.
- [84] Dalvi-Isfahan, M.; Havet, M.; Hamdami, N.; Le-Bail, A. Recent Advances of High Voltage Electric Field Technology and Its Application in Food Processing: A Review with a Focus on Corona Discharge and Static Electric Field. *J. Food Eng.* **2023**, *353*, 111551. DOI: [10.1016/j.jfoodeng.2023.111551](https://doi.org/10.1016/j.jfoodeng.2023.111551).
- [85] Lu, X.; An, S. Overview and Research Application of High Voltage Electric Field Drying Technology. *Energy Conserv.* **2022**, *41*, 75–80. DOI: [10.3969/j.issn.1004-7948.2022.06.020](https://doi.org/10.3969/j.issn.1004-7948.2022.06.020).
- [86] Qi, M.; Liu, Q.; Shi, S.; Xian, Y.; Yuan, Y. Recent Progress in the Application of High-Voltage Electric Field Technology in Food Sterilization. *Food Sci.* **2022**, *43*, 284–292. DOI: [10.7506/spkx1002-6630-20210313-175](https://doi.org/10.7506/spkx1002-6630-20210313-175).
- [87] Bai, Y.; Shi, H.; Yang, Y. Application of High Voltage Electric Field (HVEF) Drying Technology in Potato Chips. *J. Phys.: Conf. Ser.* **2013**, *418*, 012144. DOI: [10.1088/1742-6596/418/1/012144](https://doi.org/10.1088/1742-6596/418/1/012144).
- [88] Defraeye, T.; Martynenko, A. Electrohydrodynamic Drying of Food: New Insights from Conjugate Modeling. *J. Cleaner Prod.* **2018**, *198*, 269–284. DOI: [10.1016/j.jclepro.2018.06.250](https://doi.org/10.1016/j.jclepro.2018.06.250).
- [89] Ding, C.; Yang, M. Drying Characteristics and Mathematical Models of Chinese Wolfberry in DC High Voltage Electric Field. *Transact. Chin. Soc. Agric. Mach.* **2017**, *48*, 302–311. DOI: [10.6041/j.issn.1000-1298.2017.06.040](https://doi.org/10.6041/j.issn.1000-1298.2017.06.040).

- [90] Meng, Z.; Cui, X.; Liu, Y.; Wang, Y.; Hu, R.; Du, C.; Wang, S. Effect of Heat Pump-High Voltage Electric Field Combined Drying on Drying Characteristics of Yam Slices. *Food Ferment Indust.* **2023**, *49*, 124–132. DOI: [10.13995/j.cnki.11-1802/ts.033061](https://doi.org/10.13995/j.cnki.11-1802/ts.033061).
- [91] Li, Y. Experimental Study on the Drying of *Gastrodia Elata*. *Rural Pract. Technol.* **2021**, 112–113.
- [92] Taghian Dinani, S.; Hamdami, N.; Shahedi, M.; Havet, M.; Queveau, D. Influence of the Electrohydrodynamic Process on the Properties of Dried Button Mushroom Slices: A Differential Scanning Calorimetry (DSC) Study. *Food Bioprod. Process.* **2015**, *95*, 83–95. DOI: [10.1016/j.fbp.2015.04.001](https://doi.org/10.1016/j.fbp.2015.04.001).
- [93] Anukiruthika, T.; Moses, J. A.; Anandharama krishnan, C. Electrohydrodynamic Drying of Foods: Principle, Applications, and Prospects. *J. Food Eng.* **2021**, *295*, 110449. DOI: [10.1016/j.jfoodeng.2020.110449](https://doi.org/10.1016/j.jfoodeng.2020.110449).
- [94] Ni, J. Effects of Different Needle Spacing and Pretreatment Methods on Drying Characteristics of Goji Berry in High Voltage Electric Field. Master Thesis, Inner Mongolia University of Technology, **2021**.
- [95] Sriariyakul, W.; Swasdisevi, T.; Devahastin, S.; Soponronnarit, S. Drying of Aloe Vera Puree Using Hot Air in Combination with Far-Infrared Radiation and High-Voltage Electric Field: Drying Kinetics, Energy Consumption and Product Quality Evaluation. *Food Bioprod. Process.* **2016**, *100*, 391–400. DOI: [10.1016/j.fbp.2016.08.012](https://doi.org/10.1016/j.fbp.2016.08.012).
- [96] Guo, X.; Wang, W.; Liu, Y.; Ran, G.; Zhang, H.; Wang, H. Effects of Different Drying Methods on Physical and Chemical Properties and Drying Energy Consumption of *Penaeus vannamei*. *Food Indus.* **2017**, *38*, 33–38.
- [97] Duan, Q.; Zhang, S.; Li, X.; Tian, X.; Wang, D. Experimental Study on Heat Pump Drying of Apple Slices. *J. Qingdao Univ.* **2018**, *33*, 80–86. DOI: [10.13306/j.1006-9798.2018.01.016](https://doi.org/10.13306/j.1006-9798.2018.01.016).
- [98] Huang, J.; Huang, Y.; Zheng, B.; Li, S. The Impact of Different Drying Methods on the Quality of Silver Ear. *J. Chin. Inst. Food Sci. Technol.* **2010**, *10*, 167–173. DOI: [10.16429/j.1009-7848.2010.02.020](https://doi.org/10.16429/j.1009-7848.2010.02.020).
- [99] Abdelkader, T. K.; Sayed, H. A. A.; Refai, M.; Ali, M. M.; Zhang, Y.; Wan, Q.; Khalifa, I.; Fan, Q.; Wang, Y.; Abdelhamid, M. A. Machine Learning, Mathematical Modeling and 4E (Energy, Exergy, Environmental, and Economic) Analysis of an Indirect Solar Dryer for Drying Sweet Potato. *Renew. Energy.* **2024**, *227*, 120535. DOI: [10.1016/j.renene.2024.120535](https://doi.org/10.1016/j.renene.2024.120535).
- [100] Dutta, P.; Das, H.; Dutta, P. P.; Kalita, P. Evaluation of an Improved Indirect Solar Dryer for Curcuma Amada without and with Stone Chips as Thermal Energy Storage: An Investigation on Kinetics, Energy, Exergy, Quality and Economic Aspects. *J. Energy Storage.* **2024**, *79*, 110199. DOI: [10.1016/j.est.2023.110199](https://doi.org/10.1016/j.est.2023.110199).
- [101] Guo, J.; Zheng, L.; Li, Z. Effect and Interrelationship of Different Influencing Factors on Drying Performance and Energy Analysis of Cow Manure in Microwave Drying at Pilot Scale. *J. Cleaner Prod.* **2023**, *413*, 137407. DOI: [10.1016/j.jclepro.2023.137407](https://doi.org/10.1016/j.jclepro.2023.137407).
- [102] Darvishi, H.; Hadi Khoshtaghaza, M.; Najafi, G.; Zarein, M. Characteristics of Sunflower Seed Drying and Microwave Energy Consumption. *Int. Agrophys.* **2013**, *27*, 127–132. DOI: [10.2478/v10247-012-0077-8](https://doi.org/10.2478/v10247-012-0077-8).
- [103] Ning, X.; Han, C. Drying Characteristics and Quality of Taegeuk Ginseng (*P Anax Ginseng* C. A. M Eyer) Using Far-Infrared Rays. *Int. J. Food Sci. Tech.* **2013**, *48*, 477–483. DOI: [10.1111/j.1365-2621.2012.03208.x](https://doi.org/10.1111/j.1365-2621.2012.03208.x).
- [104] Jibril, A. N.; Zhang, X.; Wang, S.; Bello, Z. A.; Henry, I. I.; Chen, K. Far-Infrared Drying Influence on Machine Learning Algorithms in Improving Corn Drying Process with Graphene Irradiation Heating Plates. *J. Food Process Eng.* **2024**, *47*, e14603. DOI: [10.1111/jfpe.14603](https://doi.org/10.1111/jfpe.14603).
- [105] İsmail, O.; Gökçe Kocabay, Ö. Vacuum Oven Drying of Mussels: Mathematical Modeling, Drying Characteristics and Kinetics Study. *BFF.* **2022**, *124*, 1238–1253. DOI: [10.1108/BFFJ-05-2021-0486](https://doi.org/10.1108/BFFJ-05-2021-0486).
- [106] Hou, Y. *Process Optimization and Quality Characteristic Analysis of Vacuum Freeze-Dried Jujube Slices*; Tarim University, **2024**.
- [107] Guo, S.; Jiang, A.; Wang, L. Optimization of Technology of Chinese Gooseberry Slices by Vacuum Freeze-Drying Based on Quality and Energy Consumption. *Food Res. Develop.* **2016**, *37*, 108–112.
- [108] Zhang, N.; Shi, L.; Qi, H.; Xie, Y.; Cai, L. Effect of Radio Frequency (RF) Drying Technology on Dehydration Rate and Energy Consumption of Australia Lignite. *Dry. Technol.* **2016**, *34*, 161–166. DOI: [10.1080/07373937.2015.1012677](https://doi.org/10.1080/07373937.2015.1012677).
- [109] Ling, Z.; Ren, G.; Duan, X.; Lu, Y.; Chen, X. Design and Analysis of Radio Frequency Dryer for Small Particles (Containing Powder). *Food Machin.* **2019**, *35*, 99–103. DOI: [10.13652/j.issn.1003-5788.2019.11.020](https://doi.org/10.13652/j.issn.1003-5788.2019.11.020).
- [110] Mirzaei-Baktash, H.; Hamdami, N.; Torabi, P.; Fallah-Joshaqani, S.; Dalvi-Isfahan, M. Impact of Different Pretreatments on Drying Kinetics and Quality of Button Mushroom Slices Dried by Hot-Air or Electrohydrodynamic Drying. *LWT.* **2022**, *155*, 112894. DOI: [10.1016/j.lwt.2021.112894](https://doi.org/10.1016/j.lwt.2021.112894).
- [111] Ning, X. Drying Features of Microwave and Far-Infrared Combination Drying on White Ginseng Slices. *Korean Soc. Food Sci. Technol.* **2019**, *28*, 1065–1072. DOI: [10.1007/s10068-018-00541-0](https://doi.org/10.1007/s10068-018-00541-0).
- [112] Liu, Z.; Yang, T.; Xu, H.; Li, X.; Chen, J.; Ren, G.; Duan, X. The Influence of Drying Methods on the Physicochemical, Function, and Antioxidant Properties of Passion Fruit Peel. *Food Machin.* **2023**, *39*, 166–174. DOI: [10.13652/j.spjx.1003.5788.2023.80068](https://doi.org/10.13652/j.spjx.1003.5788.2023.80068).
- [113] Shang, T.; Yuan, Y.; Zhao, Z.; Xu, Y. Drying Kinetics and Quality of *Scutellaria baicalensis*

- Microwave-Hot Air Combined with Drying. *Chin. Trad. Herbal Drugs*. **2023**, 54, 4501–4510. DOI: [10.7501/j.issn.0253-2670.2023.14.010](https://doi.org/10.7501/j.issn.0253-2670.2023.14.010).
- [114] Zeng, S.; Wang, B.; Lv, W.; Wu, Y. Effects of Microwave Power and Hot Air Temperature on the Physicochemical Properties of Dried Ginger (*Zingiber Officinale*) Using Microwave Hot-Air Rolling Drying. *Food Chem.* **2023**, 404, 134741. DOI: [10.1016/j.foodchem.2022.134741](https://doi.org/10.1016/j.foodchem.2022.134741).
- [115] Wang, H.; Zhang, Q.; Zang, Z.; Wang, C.; Shang, J.; Huang, X.; Wan, F. Optimization of Microwave Vacuum Drying Process of Astragalus Slices by Response Surface Methodology. *Forestry Mach. Woodwork. Equip.* **2023**, 51, 59–64 + 71. DOI: [10.13279/j.cnki.fmwe.20231204.001](https://doi.org/10.13279/j.cnki.fmwe.20231204.001).
- [116] Xu, G.; Yin, H.; He, X.; Wang, D.; Zhao, Y.; Yue, J. Optimization of Microwave Vacuum Drying of Okra and the Study of the Product Quality. *J. Food Process Eng.* **2020**, 43, e13337. DOI: [10.1111/jfpe.13337](https://doi.org/10.1111/jfpe.13337).
- [117] Cui, Z.-W.; Xu, S.-Y.; Sun, D.-W.; Chen, W. Dehydration of Concentrated Ganoderma Lucidum Extraction by Combined Microwave-Vacuum and Conventional Vacuum Drying. *Dry. Technol.* **2006**, 24, 595–599. DOI: [10.1080/07373930600626792](https://doi.org/10.1080/07373930600626792).
- [118] Huang, J.; Zhang, M. Effect of Three Drying Methods on the Drying Characteristics and Quality of Okra. *Dry. Technol.* **2016**, 34, 900–911. DOI: [10.1080/07373937.2015.1086367](https://doi.org/10.1080/07373937.2015.1086367).
- [119] Li, L.; Zhang, M.; Zhou, L. A Promising Pulse-Spouted Microwave Freeze Drying Method Used for Chinese Yam Cubes Dehydration: Quality, Energy Consumption, and Uniformity. *Dry. Technol.* **2021**, 39, 148–161. DOI: [10.1080/07373937.2019.1624564](https://doi.org/10.1080/07373937.2019.1624564).
- [120] Yan, S.; Zhao, L.; Wang, Q. Drying Characteristics and Energy Consumption Analysis of Solar Heat Pump Combined System of Wolfberry. *Acta Energeticae Solaris Sinica*. **2023**, 44, 106–112. DOI: [10.19912/j.0254-0096.tynxb.2022-1261](https://doi.org/10.19912/j.0254-0096.tynxb.2022-1261).
- [121] Gan, S. H.; Ng, M. X.; Tham, T. C.; Chua, L. S.; Aziz, R.; Baba, M. R.; Chuah Abdullah, L.; Ong, S. P.; Law, C. L. Drying Characteristics of Orthosiphon Stamineus Benth by Solar-Assisted Heat Pump Drying. *Dry. Technol.* **2017**, 35, 1755–1764. DOI: [10.1080/07373937.2016.1275673](https://doi.org/10.1080/07373937.2016.1275673).
- [122] Hu, J. Characteristics and Modeling of Heat Pump and Hot-Air Combination Drying Process of Gynuraprocumbens Leaves. *Biol. Chem. Eng.* **2020**, 6, 14–17 + 24.
- [123] Zhao, H.; Dai, J.; Qiao, L.; Zhang, L.; Wu, K. Effects of Heat Pump Microwave Combined Drying on Drying Characteristics and Quality Characteristics of *Stichopus Japonicus*. *J. Yantai Univ. (Nat. Sci. Eng. Ed.)*. **2018**, 31, 329–335. DOI: [10.13951/j.cnki.37-1213/n.2018.04.009](https://doi.org/10.13951/j.cnki.37-1213/n.2018.04.009).
- [124] Zhang, B. Walnut Drying Characteristics and Quality Changes When Subjected to Combined Radio Frequency and Hot Air Heating. Master Thesis, Northwest Agriculture and Forestry University, **2018**.
- [125] Liu, Z.; Wu, G.; De, X.; Bao, X. Effect of Infrared Hot Air Combined with Drying on Nutrient Composition of Alfalfa. *Inner Mongolia Sci. Technol. Econ.* **2022**, 12, 99–103.
- [126] Jiang, D.; Xiao, H.; Zheng, Z. Effects of Different Drying Methods on Drying Characteristics, Microstructure, Quality, and Energy Consumption of *Panax Notoginseng* Roots (Araliaceae). *Dry. Technol.* **2022**, 40, 1247–1261. DOI: [10.1080/07373937.2020.1863978](https://doi.org/10.1080/07373937.2020.1863978).
- [127] Peng, J.; Wang, W.; Tang, D.; Wen, J.; Li, Yang, W.; Wu, J.; Yu, Y. Effect of Multi-Stage Far-Infrared Radiation-Assisted Heat Pump Drying on the Quality Characteristics of Longan. *Food Sci.* **2020**, 41, 118–123. DOI: [10.7506/spkx1002-6630-20191020-210](https://doi.org/10.7506/spkx1002-6630-20191020-210).
- [128] Pan, X.; Xu, B.; Zhang, J.; Wang, P. Drying Effects of Far-Infrared Radiation Heating Assisted Pulsed Vacuum Drying and Traditional Drying Methods on *Lycium barbarum*: A Comparative Analysis. *Modern Chin. Med.* **2022**, 24, 505–509. DOI: [10.13313/j.issn.1673-4890.20210224002](https://doi.org/10.13313/j.issn.1673-4890.20210224002).
- [129] Wang, T.; Ying, X.; Zhang, Q.; Xu, Y.; Jiang, C.; Shang, J.; Zang, Z.; Wan, F.; Huang, X. Drying Kinetics Prediction and Quality Effect of Ultrasonic Synergy Vacuum Far-Infrared Drying of *Codonopsis Pilosula*. *J. Food Sci.* **2024**, 89, 966–981. DOI: [10.1111/1750-3841.16890](https://doi.org/10.1111/1750-3841.16890).
- [130] Long, Y.; Zhang, M.; Mujumdar, A. S.; Chen, J. Valorization of Turmeric (*Curcuma Longa* L.) Rhizome: Effect of Different Drying Methods on Antioxidant Capacity and Physical Properties. *Dry. Technol.* **2022**, 40, 1609–1619. DOI: [10.1080/07373937.2022.2032135](https://doi.org/10.1080/07373937.2022.2032135).
- [131] Yao, N. Effects of Pretreatment and Drying Methods on Drying Characteristics and Quality of *Hippophae Rhamnoides* L. Master Thesis, Xinjiang Agricultural University, **2022**.
- [132] Zhang, F.; Xin, L.; Li, L.; Zhou, J.; Li, Z.; Lin, Y. Characteristics and Quality Analysis of Main Roots of *Panax Notoginseng* (Burk.) F. H. Chen by Microwave Vacuum-Hot Air Segmented Drying. *Jiangsu J. Agricult. Sci.* **2022**, 38, 214–222. DOI: [10.3969/j.issn.1000-4440.2022.01.026](https://doi.org/10.3969/j.issn.1000-4440.2022.01.026).
- [133] Zhang, Z.; Zhu, C.; Li, D.; Liu, C.; Jiang, N.; Wang, X.; Nie, M. Effect of Different Drying Methods on the Polyphenols and Antioxidant Activities of Burdock Roots Slices. *Jiangsu J. Agricult. Sci.* **2018**, 34, 172–178. DOI: [10.3969/j.issn.1000-4440.2018.01.025](https://doi.org/10.3969/j.issn.1000-4440.2018.01.025).
- [134] Song, S.; Huang, X.; Liu, Y.; Zhang, Q. Drying Characteristics and Quality Analysis of Hot Air-Assisted Radio Frequency and Hot-Air Drying of Jujube (*Zizyphus Jujube* Miller Cv. Jinsixiaozao). *Eng. Agric.* **2022**, 42. DOI: [10.1590/1809-4430-eng.agric.v42n1e20210112/2022](https://doi.org/10.1590/1809-4430-eng.agric.v42n1e20210112/2022).
- [135] Xu, H.; Wu, M.; Wang, Y.; Wei, W.; Sun, D.; Li, D.; Zheng, Z.; Gao, F. Effect of Combined Infrared and Hot Air Drying Strategies on the Quality of *Chrysanthemum* (*Chrysanthemum Morifolium* Ramat.) Cakes: Drying Behavior, Aroma Profiles

- and Phenolic Compounds. *Foods*. **2022**, *11*, 2240. DOI: [10.3390/foods11152240](https://doi.org/10.3390/foods11152240).
- [136] Duan, X.; Song, Y.; Chen, J.; Li, L.; Liu, W.; Cao, W.; Ren, G.; Wang, Z. Effects of Different Drying Methods on Quality Characteristics and Volatile Components of Male Flowers of *Eucommia Ulmoides*. *Transact. Chin. Soc. Agric. Eng.* **2023**, *1*–13. DOI: [10.11975/j.issn.1002-6819.202306154](https://doi.org/10.11975/j.issn.1002-6819.202306154).
- [137] Wang, Z.; Lu, F.; Zhu, G.; Yang, H.; Guo, W.; Zhai, C.; Han, J.; Wang, T.; Dong, J.; Dong, T. Characteristics and Energy Consumption of Microwave Coupled with Hot Air Drying Process of Peanut Pods. *Chin. J. Oil Crop Sci.* **2020**, *42*, 140–146. DOI: [10.19802/j.issn.1007-9084.2019142](https://doi.org/10.19802/j.issn.1007-9084.2019142).
- [138] Li, X.; Yang, L.; Wang, J.; Li, M. Study on Process Optimization and Kinetic Model of Microwave Air Combined Drying for *Phyllanthus emblica* L. *Food Ferment Indus.* **2023**, *49*, 131–139. DOI: [10.13995/j.cnki.11-1802/ts.033423](https://doi.org/10.13995/j.cnki.11-1802/ts.033423).
- [139] Wang, D. Effect of Different Drying Methods and Blanching Approaches on the Quality and Antioxidant Properties of Okra. Master Thesis, Nanjing Normal University, **2020**.
- [140] Zou, X.; Deng, H.; Wu, J.; Ji, W.; Fu, Y.; Zhang, S.; Liu, X. Evaluation of the Effects of Three Drying Methods on Physicochemical Properties of *Dendrobium Officinale* Polysaccharides Based on Coefficient of Variation Weighting Method. *J. Guangd. Pharma. Univ.* **2023**, *39*, 87–92. DOI: [10.16809/j.cnki.2096-3653.2023050402](https://doi.org/10.16809/j.cnki.2096-3653.2023050402).
- [141] Wang, Z.; Ma, L.; Lin, H.; Nie, S.; Yin, J. Effects of Drying Methods on Structural Characteristics and Antioxidant Properties of Okra Polysaccharides. *Chin. Inst. Food Sci. Technol.* **2022**, *22*, 310–318. DOI: [10.16429/j.1009-7848.2022.02.033](https://doi.org/10.16429/j.1009-7848.2022.02.033).
- [142] Liu, J.; Liu, Y.; Zhang, Q. Effect of Different Drying Methods on Activities of Polysaccharides and Its Application Extracted from *Cornus officinalis*. *Food Indus.* **2018**, *39*, 190–194.
- [143] Chen, H.; Zhang, L. Effects of Different Drying Methods on Quality of Chinese Yam. *Food Indus.* **2020**, *41*, 195–199.
- [144] Chen, H.; Gu, R.; Sun, D.; Tang, Y.; Zhang, J.; Wu, Y.; Liu, E. Effects of Different Drying Pretreatment on Antioxidant of *Arctium lappa* L. Polyphenols and Polysaccharides. *Farm Prod. Process.* **2023**, *570*, 48–52. DOI: [10.16693/j.cnki.1671-9646\(X\).2023.02.039](https://doi.org/10.16693/j.cnki.1671-9646(X).2023.02.039).
- [145] Chen, H.; Jiang, Y.; Xu, F.; Zhang, L.; Hu, J.; Wu, J.; Wang, H. Effects of Different Drying Methods on Physicochemical Properties and Antioxidant Activities of *Ligusticum Chuanxiong* Polysaccharide. *Chin. Trad. Patent Med.* **2021**, *43*, 173–177. DOI: [10.3969/j.issn.1001-1528.2021.01.034](https://doi.org/10.3969/j.issn.1001-1528.2021.01.034).
- [146] Wu, B.; Qiu, C.; Guo, Y.; Zhang, C.; Li, D.; Gao, K.; Ma, Y.; Ma, H. Comparative Evaluation of Physicochemical Properties, Microstructure, and Antioxidant Activity of Jujube Polysaccharides Subjected to Hot Air, Infrared, Radio Frequency, and Freeze Drying. *Agricult. London.* **2022**, *12*, 1606. DOI: [10.3390/agriculture12101606](https://doi.org/10.3390/agriculture12101606).
- [147] Duan, M.; Liu, K.; Shang, H.; Yang, J.; Li, R.; Wu, H. Effects of Drying Methods on Physicochemical Properties and Antioxidant Activities of Polysaccharides from *Symphytum pfficinale* L. *J. Northwest A&F Univ. (Nat. Sci. Ed.)*. **2018**, *46*, 8–14. DOI: [10.13207/j.cnki.jnwafu.2018.10.002](https://doi.org/10.13207/j.cnki.jnwafu.2018.10.002).
- [148] Li, Y.; Niu, J.; Cao, R.; Si, X.; Bian, T.; Xing, E.; Zhang, A.; Yan, X. Study on the Effect of Different Drying Processes on the Content of *Hedysarum Polybotrys* Polysaccharides. *Lishizhen Med. Mater. Medica Res.* **2019**, *30*, 92–94. DOI: [10.3969/j.issn.1008-0805.2019.01.027](https://doi.org/10.3969/j.issn.1008-0805.2019.01.027).
- [149] Dai, Y.; Wang, D.; Qing, P.; Wu, H.; Zong, Ruiye.; Ma, S.; Guo, X.; Wang, Z. Study on the Characteristics, Quality and Color Protection Application of *Dendrobium Huoshanense* Flower during Hot-Air Drying Processes. *Food Ferment. Indus.* **2021**, *47*, 193–197. DOI: [10.13995/j.cnki.11-1802/ts.024544](https://doi.org/10.13995/j.cnki.11-1802/ts.024544).
- [150] Xiao, Y.; Yang, H.; Li, D.; Gu, Q.; Liu, C.; Xu, Y.; Song, J.; Liu, C.; Zhang, Z. Effects of Different Color Protection Pretreatment on Hot Air Drying Characteristics and Quality Changes of Iron Rod Yam. *Jiangsu Agric. Sci.* **2020**, *48*, 218–223. DOI: [10.15889/j.issn.1002-1302.2020.20.042](https://doi.org/10.15889/j.issn.1002-1302.2020.20.042).
- [151] Qiu, S.; Wang, Y.; Jiang, C.; He, D.; Luo, Z. Effects of Different Color Fixative Technologies on Quality of Dried Longan. *Guangdong Agric. Sci.* **2015**, *42*, 77–83. DOI: [10.16768/j.issn.1004-874x.2015.19.005](https://doi.org/10.16768/j.issn.1004-874x.2015.19.005).
- [152] Meng, X.; Dang, B.; Zhang, W.; Zhang, J.; Dong, H.; Bai, K. Effects of Different Drying Methods on Nutritional Components and Antioxidant Activity of Nettle Leaves. *Food Indus.* **2023**, *44*, 161–166.
- [153] Gao, P.; Zhao, M.; Ma, Z.; Jiang, T.; Liu, L.; Wen, C. Effects of Different Drying Methods on Total Phenolic Content and Antioxidant Capacity of Dandelion Leaf. *Modern Food.* **2023**, *29*, 157–159. DOI: [10.16736/j.cnki.cn41-1434/ts.2023.19.037](https://doi.org/10.16736/j.cnki.cn41-1434/ts.2023.19.037).
- [154] Tan, S.; Peng, S.; Li, W.; Ren, S.; Wang, J.; Zhang, Y.; Shi, S. Effects of Different Drying Methods on Polyphenol Profile and Antioxidant Activities in Longan (*Ficus carica* Linn.). *J. Fruit Sci.* **2021**, *38*, 411–420. DOI: [10.13925/j.cnki.gsxb.20200488](https://doi.org/10.13925/j.cnki.gsxb.20200488).
- [155] Li, Q.; Lv, Y.; Yu, D.; Li, R.; Luo, S.; Yang, C. Effects of Different Drying Methods on Texture Properties, Active Components and Antioxidant Capacity of Okra. *Food Sci. Technol.* **2022**, *47*, 79–84. DOI: [10.13684/j.cnki.spkj.2022.11.018](https://doi.org/10.13684/j.cnki.spkj.2022.11.018).
- [156] Guo, H. Effects of Different Drying Methods on Polyphenol Acids of *Salvia Miltiorrhiza*. *World Latest Med. Inf.* **2015**, *15*, 107–112. DOI: [10.3969/j.issn.1671-3141.2015.17.082](https://doi.org/10.3969/j.issn.1671-3141.2015.17.082).
- [157] Zou, R.; You, Y.; Chen, Z.; Hu, K.; Ran, L. Effects of Drying Methods on Polyphenol Compounds and Antioxidant Activities of *Lonicera Japonica* Flower. *Food Sci.* **2016**, *37*, 78–83. DOI: [10.7506/spkx1002-6630-201605015](https://doi.org/10.7506/spkx1002-6630-201605015).
- [158] Zhang, Y.; Niu, Q.; Wang, T. Study on the Effect of Different Drying Processing Techniques on the Retention of Nutrient Components in Raspberry. *Modern Food.* **2023**, *29*, 175–177. DOI: [10.16736/j.cnki.cn41-1434/ts.2023.18.053](https://doi.org/10.16736/j.cnki.cn41-1434/ts.2023.18.053).

- [159] Sheng, J.; Chen, K.; Tang, Y.; Wang, X.; He, X.; Zhou, Z.; Ling, D.; Lei, Y.; Li, L.; Yi, P.; et al. Physicochemical Properties and Volatile Composition Analysis of *Jasminum sambac* (L.) Ait. Dried by Different Methods. *Modern Food Sci. Technol.* **2022**, *38*, 247–259 + 34. DOI: [10.13982/j.mfst.1673-9078.2022.8.1033](https://doi.org/10.13982/j.mfst.1673-9078.2022.8.1033).
- [160] Ning, X.; Lee, J.; Han, C. Drying Characteristics and Quality of Red Ginseng Using Far-Infrared Rays. *J. Ginseng Res.* **2015**, *39*, 371–375. DOI: [10.1016/j.jgr.2015.04.001](https://doi.org/10.1016/j.jgr.2015.04.001).
- [161] Wu, X.; Chen, G.; Hou, S.; Jia, B.; Ji, Z.; Li, L.; Xu, Q.; Chen, X.; Qian, C. Effects of Drying Process on the Polyphenol Ingredients and Antioxidant of *Oenanthe javanica* D. C. *J. Yangzhou Univ. (Agric. Life Sci. Ed.)*. **2020**, *41*, 90–95. DOI: [10.16872/j.cnki.1671-4652.2020.06.016](https://doi.org/10.16872/j.cnki.1671-4652.2020.06.016).
- [162] Deng, J.; Li, W.; Yu, L.; Chen, J.; Li, H.; Xia, C.; Tian, H.; Li, Z. Effect of Drying Temperature on Phenolic Contents and Its Antioxidant Activity of *Phyllanthus emblica* L. *Sci. Technol. Food Indus.* **2019**, *40*, 57–61. DOI: [10.13386/j.issn1002-0306.2019.24.010](https://doi.org/10.13386/j.issn1002-0306.2019.24.010).
- [163] Shen, Z.; Hou, Y.; Zhao, Y.; Zhou, H.; Chen, K.; Guo, M. Research Progress on Extraction Technology and Pharmacological Action of Flavonoids from Chinese Herbal Medicine. *Study Trace Element. Health.* **2023**, *40*, 66–68.
- [164] Liu, C.; Wang, H.; Li, A.; Tang, M.; Liu, C.; Zhang, Z.; Jiang, N.; Niu, L.; Cui, L.; Yuan, C. Effects of Different Drying Methods on the Quality and in Vitro Antioxidant Capacity of Okra Flowers. *Jiangsu J. Agric. Sci.* **2018**, *34*, 913–920. DOI: [10.3969/j.issn.1000-4440.2018.04.028](https://doi.org/10.3969/j.issn.1000-4440.2018.04.028).
- [165] Guo, X.; Fu, M.; Du, J.; Wang, X. Effects of Different Drying Methods on Antioxidant Activity and Chemical Composition of *Nelumbo Nucifera* Gaertn. *Food Ferment. Indus.* **2010**, *36*, 145–149. DOI: [10.13995/j.cnki.11-1802/ts.2010.02.026](https://doi.org/10.13995/j.cnki.11-1802/ts.2010.02.026).
- [166] Li, P.; Chen, F.; Li, X.; Zhai, J.; Wang, Z. The Effects of Different Drying Methods on Nutrient Content and Antioxidant Activity of *Paeonia ostii* 'Fengdan' flowers. *Hubei Agric. Sci.* **2021**, *60*, 111–115 + 129. DOI: [10.14088/j.cnki.issn0439-8114.2021.17.023](https://doi.org/10.14088/j.cnki.issn0439-8114.2021.17.023).
- [167] Zuo, Y.; Xu, X.; Li, Q.; Wang, M.; Meng, X.; Yan, P.; Jin, C.; Wu, D. Effects of Different Drying Methods on the Chemical Composition and Antioxidant Activity of *Albizia flos*. *Nat. Product Res. Develop.* **2024**, *36*, 125–136. DOI: [10.16333/j.1001-6880.2024.1.015](https://doi.org/10.16333/j.1001-6880.2024.1.015).
- [168] Qiu, C.; Kong, H.; Zheng, F.; Xu, K.; Ma, G.; Chen, J. Effects of Different Drying Methods on Flavonoids Content and Antioxidant Capacity of *Dendrobium Candidum* Flower. *Lishizhen Med. Mater. Medica Res.* **2020**, *31*, 598–600. DOI: [10.3969/j.issn.1008-0805.2020.03.026](https://doi.org/10.3969/j.issn.1008-0805.2020.03.026).
- [169] Mao, Y.; Jin, X.; Liu, X.; Zhang, Y.; Wang, X. Effects of Different Drying Methods on the Quality of Lanzhou Lily. *Chin. Arch. Trad. Chin. Med.* **2023**, 1–10.
- [170] Cheng, L.; Liu, Z.; Piao, M. Effects of Drying Methods on the Antioxidant Activity of Flavonoids Extracted from the Leaves of *Angelica Keiskei koidzumii*. *J. Qingdao Agric. Univ. (Nat. Sci.)*. **2018**, *35*, 138–143.
- [171] Bai, Y.; Duan, J.; Su, S.; Qian, Y.; Qian, D.; Ouyang, Z. Analysis of Dynamic Changes of Flavonoids and Alkaloids During Different Drying Process of *Morus Alba* Leaves. *Zhong Yao Cai*. **2014**, *37*, 1158–1163. DOI: [10.13863/j.issn1001-4454.2014.07.012](https://doi.org/10.13863/j.issn1001-4454.2014.07.012).
- [172] Wang, M.; Mu, L.; Zhou, Z.; Jiang, H. Effects of Different Preliminary Treatments on Total Flavonoids Extraction from *Elaeagnus angustifolia* L. leaves. *China Brewing*. **2013**, *32*, 102–105.
- [173] Wang, Y.; Jia, X.; Wang, Y.; Zhou, J.; Wang, Li, X.; Zhang, J.; Xu, Q.; Shi, W.; Hu, F. Effects of Different Drying Methods on Flavonoids Content and Antioxidant Activity of *Epimedium Folium*. *Chin. Trad. Herbal Drugs*. **2021**, *52*, 4193–4200. DOI: [10.7501/j.issn.0253-2670.2021.14.009](https://doi.org/10.7501/j.issn.0253-2670.2021.14.009).
- [174] He, H.; Xiao, S.; Tian, Z.; Zeng, Y.; Zhao, R.; Peng, Y.; Xiao, P. Different Drying Methods for Flavonoids in *Scutellaria baicalensis* Leaves Study on the Effects of Components. *Modern Chin. Med.* **2012**, *14*, 31–35. DOI: [10.13313/j.issn.1673-4890.2012.12.012](https://doi.org/10.13313/j.issn.1673-4890.2012.12.012).
- [175] Su, D. Effect of Drying Method on Flavonoid Content of *Ginkgo Biloba* Leaves. *Modern Agric. Sci. Technol.* **2009**, *14*, 78–79.
- [176] Hong, C.; Shen, D.; Wei, B.; Zhang, J. Effects of Drying and Storage Methods on the Content of Total Flavonoids in the Root of *Tuber Trifolium*. *J. Zhejiang Agric. Sci.* **2019**, *60*, 2316–2318. DOI: [10.16178/j.issn.0528-9017.20191242](https://doi.org/10.16178/j.issn.0528-9017.20191242).
- [177] Chang, F.; Duan, X.; Wang, Q.; Yang, X.; Yang, J.; Wang, D.; Wu, Y. Effect of Different Drying Methods on Flavones Content in *Pueraria*. *Sci. Technol. Food Indus.* **2014**, *35*, 78–81 + 86. DOI: [10.13386/j.issn1002-0306.2014.09.007](https://doi.org/10.13386/j.issn1002-0306.2014.09.007).
- [178] Wu, L.; Wei, L.; Wei, J.; Zhong, W.; Tan, Y. Study on Quality Analysis of *Panax Notoginseng* Based on Composition. *Hubei Agric. Sci.* **2022**, *61*, 195–198. DOI: [10.14088/j.cnki.issn0439-8114.2022.15.036](https://doi.org/10.14088/j.cnki.issn0439-8114.2022.15.036).
- [179] Zang, Z.; Huang, X.; Ma, G.; Zhang, Q.; Jiang, C.; Wan, F. Evaluation of Drying Characteristics and Physicochemical Properties of *Angelica Sinensis* Radix under Different Drying Methods Based on Combination Entropy Weight and Variable Coefficient Method. *Chin. Trad. Herbal Drugs*. **2022**, *53*, 7403–7413. DOI: [10.7501/j.issn.0253-2670.2022.23.010](https://doi.org/10.7501/j.issn.0253-2670.2022.23.010).
- [180] Wu, L.; Lu, Y.; Li, H.; Qian, D.; Duan, L. X.; Guo, S.; Duan, J. Analysis and Evaluation of Different Drying Methods for *Lycii Fructus* Based on Multi-Type Functional Components. *Chin. Trad. Herbal Drugs*. **2022**, *53*, 2125–2136. DOI: [10.7501/j.issn.0253-2670.2022.07.023](https://doi.org/10.7501/j.issn.0253-2670.2022.07.023).
- [181] Shi, X.; Wang, L.; Wang, Y.; Qi, Y.; Liu, H.; Li, J.; Zhang, Y. The Effects of Four Different Drying Methods on Total Flavonoids and Total Organic

- Acid Contents in Hawthorn. *Shandong Sci.* **2018**, 31, 14–19. DOI: [10.3976/j.issn.1002-4026.2018.05.003](https://doi.org/10.3976/j.issn.1002-4026.2018.05.003).
- [182] Guo, X.; Chen, J.; Tang, H.; Du, Y.; Zhu, X.; Liu, L. Study on Extraction Methods and Technology of Sapomin in Chinese Herbal Medicine. *J. Inner Mongolia Med. Univ.* **2013**, 35, 116–119. DOI: [10.16343/j.cnki.issn.2095-512x.2013.s1.053](https://doi.org/10.16343/j.cnki.issn.2095-512x.2013.s1.053).
- [183] Zhu, M.; Sun, Y.; Wang, Y.; Bai, H.; Yang, B.; Wang, Q.; Kuang, H. Research Progress on the Anticancer Effect and Mechanism of Chinese Herbal Medicine Saponins. *Chin. J. Experiment. Trad. Med. Formul.* **2024**, 1–12. DOI: [10.13422/j.cnki.syfjx.20240426](https://doi.org/10.13422/j.cnki.syfjx.20240426).
- [184] Zhang, J.; Ding, B.; Zhang, H.; Qi, J.; Shen, Y.; Zhou, N.; Chen, X.; Yang, D. Effects of Different Drying Methods on Total Saponin Content and Antioxidant Capacity of Paris Polyphylla Var. yunnanensis. *Chin. J. Inform. Trad. Chin. Med.* **2016**, 23, 95–97. DOI: [10.3969/j.issn.1005-5304.2016.07.024](https://doi.org/10.3969/j.issn.1005-5304.2016.07.024).
- [185] Wang, F.; Shang, Z.; Ma, Z.; Liu, S.; Wang, Z.; Zhang, J.; Lu, J. Influence of Different Drying Methods on Steroidal Saponins and Homoisoflavonoids in Ophiopogon Japonicus. *Zhongguo Zhong Yao Za Zhi*. **2016**, 41, 4393–4399. DOI: [10.4268/cjcm20162317](https://doi.org/10.4268/cjcm20162317).
- [186] Li, F.; Li, Q.; Song, D.; Liu, P.; Lv, C.; Wang, J.; Jia, L.; Lu, J. Effect of Different Drying Methods on Ginsenosides in Flower of Panax Ginseng and Panax Quinquefolius. *Chin. Trad. Herbal Drugs*. **2015**, 46, 2937–2942. DOI: [10.7501/j.issn.0253-2670.2015.19.020](https://doi.org/10.7501/j.issn.0253-2670.2015.19.020).
- [187] Xie, Q.; Chen, Z.; Pan, Y.; Yang, J. Different Dry Means on the Contents of Total Saponins and Esculentoside A of Phytolacca Americana. *Hubei Agricult. Sci.* **2019**, 58, 126–128, 132. DOI: [10.14088/j.cnki.issn0439-8114.2019.08.029](https://doi.org/10.14088/j.cnki.issn0439-8114.2019.08.029).
- [188] Peng, L.; Yang, B.; Cheng, H.; Zhang; Sun, T.; Zhang, M.; Yang, X.; Liu, A.; Yan, Y.; Hu, B. Effects of Different Drying Methods on Active Constituents of Root Bark and Root of Polygala Tenuifolia. *Chin. Trad. Herbal Drugs*. **2018**, 49, 5010–5017. DOI: [10.7501/j.issn.0253-2670.2018.21.009](https://doi.org/10.7501/j.issn.0253-2670.2018.21.009).
- [189] Lai, W.; Yang, L.; Han, J.; Liu, F.; Ma, J.; Cai, E.; Han, M. Effects of Different Drying Methods on the Quality of Ginseng in Forest. *Chin. Wild Plant Res.* **2023**, 42, 62–68. DOI: [10.3969/j.issn.1006-9690.2023.02.011](https://doi.org/10.3969/j.issn.1006-9690.2023.02.011).
- [190] Jiao, J.; Wang, Y.; Xiong, Y.; Yang, Y.; Wu, Z.; Xiong, Y.; Yang, M. Classification of Chinese Medicines Containing Volatile Oil and Analysis of Quality Influencing Factors Based on First Part in Chinese Pharmacopoeia 2015 Edition. *Chin. J. Experim. Tradit. Med. Form.* **2019**, 25, 197–206. DOI: [10.13422/j.cnki.syfjx.20190322](https://doi.org/10.13422/j.cnki.syfjx.20190322).
- [191] Xue, M.; Qiao, J.; Wang, Q.; Xu, Y.; Wu, Q.; Dai, S. Effects of Different Drying Methods on Quality and Antibacterial Activity of Essential Oils from Eupatorium Fortunei. *Chin. Tradit. Patent Med.* **2018**, 40, 2249–2254. DOI: [10.3969/j.issn.1001-1528.2018.10.027](https://doi.org/10.3969/j.issn.1001-1528.2018.10.027).
- [192] Ouyang, T.; Yang, Q.; Huang, X.; Yan, H.; Liu, D.; Liu, S. Effects of Different Drying Methods on Composition and Antioxidative Activities of Volatile Oils from Cymbopogon Citrates. *Chin. J. Inform. Trad. Chin. Med.* **2016**, 23, 99–102. DOI: [10.3969/j.issn.1005-5304.2016.11.024](https://doi.org/10.3969/j.issn.1005-5304.2016.11.024).
- [193] Fang, L. Effects of Extracting Process of Volatile Oil and the Content Assay of Germacrone with Different Drying Methods. *Strait Pharma. J.* **2009**, 21, 38–40.
- [194] Zheng, D.; Han, H.; Zhou, Y.; Xu, X.; Pan, X.; Liao, H. Determination of 8 Active Components in Volatile Oil of Corolla Alba with Different Drying Methods by GC. *Lishizhen Med. Mater. Medica Res.* **2019**, 30, 2895–2898.
- [195] Zhou, B.; Liu, P.; Lu, X.; Qian, D.; Duan, J. Optimization of Suitable Drying Processing Methods of Angelica Dahurica. *Zhong Yao Cai*. **2016**, 39, 1509–1515. DOI: [10.13863/j.issn1001-4454.2016.07.016](https://doi.org/10.13863/j.issn1001-4454.2016.07.016).